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SURGICAL ROBOT SYSTEMS COMPRISING ROBOTIC TELEMANIPULATORS AND INTEGRATED LAPAROSCOPY

Abstract

Surgical robot systems for remote manipulation having robotic telemanipulators are provided. The surgical robot systems are well adapted for use by the surgeon, seamlessly integratable into the operation room, allow for a surgeon to work between the robot and the patient throughout a surgery in a sterile manner, are relatively low cost, and/or permit integrated laparoscopy. The system preferably includes a master console having a plurality of master links interconnected by a plurality of master joints, and a handle coupled to the master console for operating the telemanipulator. The system further includes a slave console operatively coupled to the master console and having a plurality of slave links interconnected by a plurality of slave joints that move responsive to movement at the master console to permit an end-effector to perform surgery.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International PCT Patent application Serial No. PCT/IB2020/050039, filed Jan. 4, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/505,585, filed Jul. 8, 2019, which is a continuation of U.S. patent application Ser. No. 16/269,383, filed Feb. 6, 2019, now U.S. Pat. No. 10,413,374, which claims the benefit of priority of U.S. Provisional patent application Ser. No. 62/788,781, filed Jan. 5, 2019, and U.S. Provisional patent application Ser. No. 62/627,554, filed Feb. 7, 2018, the entire contents of each of which are incorporated herein by reference.

International PCT Patent application Serial No. PCT/IB2020/050039 is a continuation-in-part of PCT/IB2019/050961, filed Feb. 6, 2019, published as WO 2019/155383, which claims the benefit of priority of U.S. Provisional patent application Ser. No. 62/788,781, filed Jan. 5, 2019, and U.S. Provisional patent application Ser. No. 62/627,554, filed Feb. 7, 2018, the entire contents of each of which are incorporated herein by reference.

FIELD OF USE

[0002] This application generally relates to remotely actuated surgical robot systems having robotic telemanipulators.

BACKGROUND

[0003] Numerous environments and applications call for remote actuation with teleoperated surgical devices. These applications include the ability to perform fine manipulation, to manipulate in confined spaces, manipulate in dangerous or contaminated environments, in clean-room or sterile environments and in surgical environments, whether open field or minimally invasive. While these applications vary, along with parameters such as precise tolerances and the level of skill of the end user, each demands many of the same features from a teleoperated system, such as the ability to carry out dexterous manipulation with high precision.

[0004] Surgical applications are discussed in the following disclosure in more detail as exemplary of applications for a teleoperated device system where known devices exist but significant shortcomings are evident in previously-known systems and methods.

[0005] Open surgery is still the preferred method for many surgical procedures. It has been used by

the medical community for many decades and typically required making long incisions in the abdomen or other area of the body, through which traditional surgical tools are inserted. Due to such incisions, this extremely invasive approach results in substantial blood loss during surgery and, typically, long and painful recuperation periods in a hospital setting.

[0006] Laparoscopy, a minimally invasive technique, was developed to overcome some of the disadvantages of open surgery. Instead of large through-wall incisions, several small openings are made in the patient through which long and thin surgical instruments and endoscopic cameras are inserted. The minimally invasive nature of laparoscopic procedures reduces blood loss and pain and shortens hospital stays. When performed by experienced surgeons, a laparoscopic technique can attain clinical outcomes similar to open surgery. However, despite the above-mentioned advantages, laparoscopy requires a high degree of skill to successfully manipulate the rigid and long instrumentation used in such procedures. Typically, the entry incision acts as a point of rotation, decreasing the freedom for positioning and orientating the instruments inside the patient. The movements of the surgeon's hand about this incision point are inverted and scaled-up relative to the instrument tip ("fulcrum effect"), which reduces dexterity and sensitivity and magnifies any tremors of the surgeon's hands. In addition, the long and straight instruments force the surgeon to work in an uncomfortable posture for hands, arms and body, which can be tremendously tiring during a prolonged procedure. Therefore, due to these drawbacks of laparoscopic instrumentation, minimally invasive techniques are mainly limited to use in simple surgeries, while only a small minority of surgeons is able to use such instrumentation and methods in complex procedures.

[0007] To overcome the foregoing limitations of previously-known systems, surgical robotic systems were developed to provide an easier-to-use approach to complex minimally invasive surgeries. By means of a computerized robotic interface, those systems enable the performance of remote laparoscopy where the surgeon sits at a console manipulating two master manipulators to perform the operation through several small incisions. Like laparoscopy, the robotic approach is also minimally invasive, providing the above-mentioned advantages over open surgery with respect to reduced pain, blood loss, and recuperation time. In addition, it also offers better ergonomics for the surgeon compared to open and laparoscopic techniques, improved dexterity, precision, and tremor suppression, and the removal of the fulcrum effect. Although being technically easier, robotic surgery still involves several drawbacks. One major disadvantage of previously-known robotic surgical systems relates to the extremely high complexity of such systems, which contain four to five robotic arms to replace the hands of both the surgeon and the assistant, integrated endoscopic imaging systems, as well as the ability to perform remote surgery, leading to huge capital costs for acquisition and maintenance, and limiting the affordability for the majority of surgical departments worldwide. Another drawback of these systems is the bulkiness of previously-known surgical robots, which compete for precious space within the operating room environment and significantly increasing preparation time. Access to the patient thus may be impaired, which raises safety concerns.

[0008] For example, the da Vinci® surgical systems (available by Intuitive Surgical, Inc., Sunnyvale, California, USA) is a robotic surgical system for allowing performance of remote laparoscopy by a surgeon. However, the da Vinci® surgical systems are very complex robotic systems, with each system costing around \$2,000,000 per robot, \$150,000 per year for servicing, and \$2,000 per surgery for surgical instruments. The da Vinci® surgical system also requires a lot of space in the operating room, making it hard to move around to a desired location within the operating room, and difficult to switch between forward and reverse surgical workspaces (also known as multi-quadrant surgery).

[0009] Moreover, as the surgeon's operating console is typically positioned away from the surgical site, the surgeon and the operating console are not in the sterile zone of the operating room. If the surgeon's operating console is not sterile, the surgeon is not permitted to attend to the patient if necessary without undergoing additional sterilization procedures. During certain surgical

operations, a surgeon may need to intervene at a moment's notice, and current bulky robotic systems may prevent the surgeon from quickly accessing the surgical site on the patient in a timely, life-saving manner.

[0010] WO97/43942 to Madhani, WO98/25666 to Cooper, and U.S. Patent Application Publication No. 2010/0011900 to Burbank each discloses a robotic teleoperated surgical instrument designed to replicate a surgeon's hand movements inside the patient's body. By means of a computerized, robotic interface, the instrument enables the performance of remote laparoscopy, in which the surgeon, seated at a console and manipulating two joysticks, performs the operation through several small incisions. Those systems do not have autonomy or artificial intelligence, being essentially a sophisticated tool that is fully controlled by the surgeon. The control commands are transmitted between the robotic master and robotic slave by a complex computer-controlled mechatronic system, which is extremely costly to produce and maintain and requires considerable training for the hospital staff.

[0011] WO2013/014621 to Beira, the entire contents of which are incorporated herein by reference, describes a mechanical teleoperated device for remote manipulation which comprises master-slave configuration including a slave unit driven by a kinematically equivalent master unit, such that each part of the slave unit mimics the movement of a corresponding part of the master unit. A typical master-slave telemanipulator provides movement in seven degrees-of-freedom. Specifically, these degrees of freedom include three translational macro movements, e.g., inward/outward, upward/downward, and left/right degrees-of-freedom, and four micro movements including one rotational degree-of-freedom, e.g., pronosupination, two articulation degrees-of-freedom, e.g., yaw and pitch, and one actuation degree-of-freedom, e.g., open/close. Although the mechanical transmission system described in that publication is well adapted to the device, the low-friction routing of the cables from handles through the entire kinematic chain to the instruments is costly, complex, bulky, and requires precise calibration and careful handling and maintenance.

[0012] In addition, previously-known purely mechanical solutions do not offer wrist alignment, low device complexity, low mass and inertia, high surgical volume, and good haptic feedback. For example, with a purely mechanical teleoperated device, in order to perform a pure pronosupination/roll movement of the instrument, the surgeon typically has to perform a combined pronosupination/roll movement of his hand/forearm as well as a translational movement on a curved path with his wrist. Such movements are complex to execute properly, and if not done properly, the end-effector pitches and yaws creating undesired parasitic movements.

[0013] Further, the routing of the articulation and actuation degrees-of-freedom cables through mechanical telemanipulators may limit the dexterity of the angular range of the various joints of the telemanipulator link-and-joint structure. This in turn limits the available surgical volume of the instruments accessible within the patient. During rapid movements of the mechanical telemanipulators, inertia of the telemanipulators also may be disturbing and result in over-shoot of the target and fatigue of the surgeon's hand. Part of this mass can be attributed to parts and components required to route the actuation and articulation degrees-of-freedom.

[0014] Accordingly, it would be desirable to provide remotely actuated surgical robot systems having robotic telemanipulators that are well adapted for use by the surgeon, seamlessly integrated into the operation room, allow for a surgeon to work between the robot and the patient in a sterile manner, are relatively low cost, and/or permit integrated laparoscopy.

[0015] It would further be desirable to provide a remotely actuated surgical robot having mechanical and/or electromechanical telemanipulators.

SUMMARY

[0016] The present invention overcomes the drawbacks of previously-known systems by providing remotely actuated surgical robot systems having robotic telemanipulators that are preferably well adapted for use by the surgeon, seamlessly integratable into the operation room, allow for a surgeon to work between the robot and the patient throughout a surgery in a sterile manner, are relatively

low cost, and/or permit integrated laparoscopy.

[0017] As will be understood by a person having ordinary skill in the art, the term “master” used herein refers to components controlled by the surgeon and may be referred to as “surgeon,” and the term “slave” used herein refers to components that interact with the patient undergoing the surgery and may be referred to as “patient.” For example, the terms “master console” and “surgeon console” are interchangeable and the terms “slave console” and “patient console” are interchangeable, etc. The surgical robot system for remote manipulation includes a master console having a plurality of master links, and a handle coupled to the master console such that movement applied at the handle moves at least one of the plurality of master links. The master console may be designed to remain sterile during the surgery. In accordance with one aspect, the handle may be removably coupled to the master console such that the handle is sterile during the surgery and sterilizable while removed for additional surgeries. For example, the handle may be removably coupled to the master console via, e.g., a clip attachment or a screw attachment. The removable handle may be purely mechanical without electronics such as circuits, sensors, or electrically coupled buttons to facilitate sterilization between surgeries while the handle is removed from the master console. In this manner, the master console may be sterile (e.g., covered with a sterile drape except at the handles) during the surgery while permitting the surgeon to have the tactile feedback available from direct contact with the robot's handles.

[0018] The surgical robot system further includes a slave console having a plurality of slave links. In accordance with one aspect, the distal end of the slave console may be rotatable about an alpha-axis of an angulation slave link of the plurality slave links such that the distal end of the slave console is positionable in a manner to permit a user to move from the master console to manually perform a laparoscopic procedure on a patient undergoing the surgery.

[0019] In addition, the system includes an end-effector coupled to the slave console, wherein the end-effector moves responsive to movement applied at the handle and responsive to movement at the slave console to perform the surgery. For example, the slave console may include a plurality of actuators, e.g., motors, operatively coupled to the end-effector that, when activated responsive to actuation at the handle, apply translational macro-movements to the plurality of slave links during a macro-synchronization state, but not in an unsynchronized macro state, and apply micro-movements to the end-effector during a micro-synchronization state, but not in an unsynchronized micro state. Moreover, the surgical robot system may include an instrument having a proximal end and a distal end, the proximal end having an instrument hub designed to be coupled to the distal end of the slave console, and the distal end having the end-effector.

[0020] The handle may include a retractable piston that moves responsive to actuation of the handle. Thus, at least one sensor of the master console is designed to sense movement of the retractable piston to cause the plurality of actuators to make corresponding micro-movements at the end-effector. In accordance with one aspect of the present invention, the slave console does not respond to movement at the master console unless the at least one sensor senses at least a predetermined amount of the retractable piston. Further, at least one sensor coupled to the handle may be designed to sense an actuation pattern of the handle that transitions the robot from an unsynchronized micro state to the micro-synchronized state. For example, in the unsynchronized micro state, movement at the handle sensed by the plurality of sensors does not cause corresponding micro-movement by the end-effector until the robot is transitioned to the micro-synchronized state because the at least one sensor senses the actuation pattern of the handle.

[0021] The master console may include a mechanical constraint designed to constrain movement of at least one master link of the plurality of master links, and may further include a clutch that when actuated prevents translational macro-movement of the plurality of master links. The surgical robot system further may include a display coupled to the master console that permits a user to visualize the end-effector during operation of the telemanipulator. Additionally, the system may include a removable incision pointer that permits alignment of the distal end of the slave console with a

trocar positioned within a patient undergoing the surgery.

[0022] Moreover, the base of the slave console may be coupled to a proximal slave link of the plurality of slave links via a proximal slave joint of a plurality of slave joints such that the plurality of slave links and joints are moveable about the proximal slave joint to position the distal end of the slave console at a desired horizontal location prior to performing the surgery while the base of the slave console remains stationary. In addition, the base of the slave console may include an adjustable vertical column coupled to the proximal slave link of the plurality of slave links. The adjustable vertical column may adjust a height of the plurality of slave links and joints to position the distal end of the slave console at a desired vertical location prior to operation of the telemanipulator.

[0023] In accordance with one aspect of the present application, slave links and joints of the pluralities of slave links and joints distal to a beta joint of the plurality of slave joints are designed to move relative to the beta joint to flip the distal end of the slave console between a forward surgical workspace and a reverse surgical workspace while slave links of the plurality of slave links proximal to the beta joint, and a base of the slave console, remain stationary.

[0024] The surgical robot system also may include a controller operatively coupled to the plurality of actuators such that the plurality of actuators apply movement to the plurality of slave links of the slave console responsive to instructions executed by the controller. For example, the controller may execute instructions to cause the plurality of actuators to move the plurality of slave links of the slave console to a home configuration where, in the home configuration, the plurality of slave links are retracted such that the end-effector is positionable within a trocar inserted in a patient undergoing the surgery. In addition, the controller may execute instructions to cause the plurality of actuators to move an angulation slave link of the plurality slave links to an angle such that the angulation slave link and the slave links of the slave console proximal to the angulation slave link remain stationary during operation of the telemanipulator. Accordingly, at the angle of the angulation slave link, the distal end of the slave console permits the end-effector to perform the surgery in a semi-spherical surgical workspace tilted at an angle essentially parallel to the angle of the angulation slave link.

[0025] In accordance with another aspect of the present invention, the master console has a master controller and the slave console has a slave controller, such that the master controller may execute instructions based on movement sensed at the handle and transmit signals to the slave controller based on the movement. Accordingly, the slave controller may receive the signals and execute instructions to move at least one of the plurality of slave links or the end-effector, or both, based on the signals transmitted from the master controller. For example, the slave console may include a right slave telemanipulator, a right slave controller, a left slave telemanipulator, and a left slave controller, and the master console may include a right master telemanipulator, a left master telemanipulator, and master controller, such that, in a forward surgical workspace configuration, the master controller communicates with the right slave controller to cause the right slave telemanipulator to move responsive to movement at the right master telemanipulator and the master controller communicates with the left slave controller to cause the left slave telemanipulator to move responsive to movement at the left master telemanipulator. Additionally, in accordance with some embodiments, in a reverse surgical workspace configuration, the master controller communicates with the left slave controller to cause the left slave telemanipulator to move responsive to movement at the right master telemanipulator and the master controller communicates with the right slave controller to cause the right slave telemanipulator to move responsive to movement at the left master telemanipulator.

[0026] Accordingly, a distal end of the right slave telemanipulator may be rotatable about the alpha-axis of a right angulation slave link of the plurality of right slave links, and a distal end of the left slave telemanipulator may be rotatable about the alpha-axis of a left angulation slave link of the plurality of left slave links such that the distal ends of the right and left slave telemanipulators are

positionable in a manner to permit a user to move from the master console to manually perform a laparoscopic procedure on a patient undergoing the surgery. In addition, the right handle may be removably coupled to the right master telemanipulator and the left handle may be removably coupled to the left master telemanipulator.

[0027] In accordance with yet another aspect of the present invention, a system for remote manipulation to perform surgery is provided. The system includes a patient console having a plurality of patient links coupled to a base, and a surgical instrument coupled to the patient console. A distal region of the surgical instrument may be inserted into a patient at a surgical site for performing robotic surgery. The system further includes a controller that executes instructions to: cause, in a surgical mode, at least one of the plurality of patient links to move responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery, and to cause the patient console to transition from the surgical mode to a laparoscopic mode where the plurality of patient links are retracted away from the patient while the base of the patient console remains stationary to expose the surgical site to permit a surgeon to perform non-robotic surgery at the surgical site without interference from the plurality of patient links.

[0028] In addition, the controller further may execute instructions to determine that the surgical instrument has been removed from the patient at the surgical site, such that the controller only cause the patient console to transition from the surgical mode to the laparoscopic mode if the surgical instrument has been removed. For example, the controller may determine that the surgical instrument has been removed from the patient by determining that the surgical instrument has been decoupled from the patient console. Moreover, the controller may cause the patient console to transition from the surgical mode to the laparoscopic mode responsive to user input received at the patient console. In addition, the handle may be removably coupled to the surgeon console such that the handle is sterile during the surgery and sterilizable while removed for additional surgeries. The system further may include a display coupled to the surgeon console to permit the surgeon to visualize the surgical instrument during operation of the system.

[0029] Further, the controller further may execute instructions to cause, in the surgical mode, the at least one of the plurality of patient links to move responsive to movement applied at the handle of the surgeon console at a scaled degree. For example, the controller may execute instructions to cause, in the surgical mode, scaled micro movements at the surgical instrument in a micro degree-of-freedom responsive to corresponding movement applied at the handle of the surgeon console. Micro movements applied at the surgical instrument may be independently scalable for each of the micro degrees-of-freedom such that the scaled micro movements in the micro degree-of-freedom are at a different scale than second scaled micro movement at the surgical instrument in a second micro degree-of-freedom. Moreover, the surgeon console may include a clutch for preventing micro-movements at the surgical instrument responsive to micro-movements applied at the handle of the surgeon console when the clutch is actuated.

[0030] In accordance with another aspect of the present invention, a method for remotely performing a surgery is provided. The method may include: coupling a surgical instrument to a patient console comprising a plurality of patient links coupled to a base; inserting a distal region of the surgical instrument into a patient at a surgical site for performing robotic surgery; moving, in a surgical mode, at least one of the plurality of patient links responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery; and transitioning the patient console from the surgical mode to a laparoscopic mode where the plurality of patient links are retracted away from the patient while the base of the patient console remains stationary to expose the surgical site to permit a surgeon to perform non-robotic surgery at the surgical site without interference from the plurality of patient links.

[0031] In accordance with yet another aspect of the present invention, another system for remote

manipulation to perform surgery is provided. The system may include a patient console having an alignment joint and a plurality of patient links coupled to a base, and a surgical instrument coupled to the patient console. A distal region of the surgical instrument may be inserted into a patient at a surgical site for performing robotic surgery. The system further may include a controller that executes instructions to: set a virtual center-of-motion based on an alignment of the alignment joint and the surgical site; and cause at least one of the plurality of patient links to move responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery, wherein movement of the surgical instrument is restricted about the virtual remote center-of-motion to maintain alignment of the patient joint with the surgical site during the surgery.

[0032] The system further may include an incision pointer that may be removably coupled to the alignment joint to permit alignment of the alignment joint and the surgical site. For example, the incision pointer may be removably coupled to the alignment joint via a magnetic attachment. In addition, the system may include a trocar positioned within the patient at the surgical site, such that the virtual center-of-motion is set based on the alignment of the alignment joint and the trocar.

[0033] In accordance with another aspect of the present invention, another method for remotely performing a surgery is provided. The method may include: aligning a patient joint of a plurality of patient joints of a patient console with a trocar insertion site, the plurality of patient joints interconnected by a plurality of patient links, the patient console operatively coupled to a surgeon console and configured to move responsive to movement applied at a handle of the surgeon console; setting a virtual remote center-of-motion based on the alignment of the patient joint with the trocar insertion site; and moving at least one of the plurality of patient links responsive to movement applied at the handle to move a surgical instrument coupled to the patient console to perform the surgery, wherein movement of the surgical instrument is restricted about the virtual remote center-of-motion to maintain alignment of the patient joint with the trocar insertion site during the surgery.

[0034] In accordance with yet another aspect of the present invention, another system for remote manipulation to perform surgery is provided. The system may include a patient console having an alignment joint and a plurality of patient links coupled to a base, and a surgical instrument coupled to the patient console. A distal region of the surgical instrument may be inserted into a patient at a surgical site for performing robotic surgery. The system further may include a controller that executes instructions to cause, in the surgical mode, scaled micro movements at the surgical instrument in micro degrees-of-freedom responsive to corresponding movement applied at the handle of the surgeon console, wherein the scaled micro movements at the surgical instrument in the micro degrees-of-freedom are greater than the corresponding movements applied at the handle of the surgeon console. Micro movements applied at the surgical instrument may be independently scalable for each of the micro degrees-of-freedom such that the scaled micro movements in a first micro degree-of-freedom are at a different scale than second scaled micro movement at the surgical instrument in a second micro degree-of-freedom.

[0035] Moreover, the surgeon console may include a clutch for preventing micro-movements at the surgical instrument responsive to micro-movements applied at the handle of the surgeon console when the clutch is actuated. For example, the surgeon may articulate the instrument end-effector into a certain position (e.g. using the roll, pitch, and/or yaw degrees-of-freedom of the end-effector) via the handle, then actuate the clutch, move the handle back into a more ergonomic position while the instrument end-effector remains stationary, and then release the clutch to continue relative micro-movements from the handle to the instrument end-effector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0036] FIG. 1 shows an exemplary remotely actuated surgical robot system having robotic telemanipulators constructed in accordance with the principles of the present invention.

[0037] FIG. 2A shows an exemplary master console constructed in accordance with the principles of the present invention.

[0038] FIG. 2B shows an exemplary display constructed in accordance with the principles of the present invention.

[0039] FIG. 2C shows another exemplary master console constructed in accordance with the principles of the present invention.

[0040] FIG. 3A shows the master console of FIG. 2A in a seated configuration, and FIGS. 3B and 3C show the master console of FIG. 2A in a standing configuration.

[0041] FIG. 4 shows an exemplary master console handle constructed in accordance with the principles of the present invention.

[0042] FIG. 5A shows an exemplary handle grip constructed in accordance with the principles of the present invention. FIGS. 5B and 5C show the handle grip of FIG. 5A removably coupling with the master console handle of FIG. 4A in accordance with the principles of the present invention.

[0043] FIGS. 5D-5F show an exemplary handle grip removably coupling with a master console handle via a clip attachment in accordance with the principles of the present invention.

[0044] FIG. 5G shows an exemplary sterile drape cap coupled to the master console handle in accordance with the principles of the present invention.

[0045] FIG. 6 shows an exemplary handle grip removably coupling with a master console handle via a screw attachment in accordance with the principles of the present invention.

[0046] FIGS. 7A-7C show actuation steps of the handle grip of FIG. 5A in accordance with the principles of the present invention.

[0047] FIGS. 7D and 7E are cross-sectional views of the handle grip of FIG. 5A coupled to the master console handle.

[0048] FIG. 8A shows another exemplary master console handle constructed in accordance with the principles of the present invention.

[0049] FIGS. 8B and 8C show movement of the handle grips of the master console handle of FIG. 8A in accordance with the principles of the present invention.

[0050] FIGS. 8D and 8E are interior views of the master console handle of FIG. 8A.

[0051] FIGS. 8F and 8G show an exemplary sterile drape interface in accordance with the principles of the present invention.

[0052] FIGS. 9A-9C show the handle grip of FIG. 8A removably coupling with the master console handle of FIG. 8A in accordance with the principles of the present invention.

[0053] FIGS. 9D-9F show the handle grip of FIG. 8A decoupling with the master console handle of FIG. 8A in accordance with the principles of the present invention.

[0054] FIGS. 10A-10C show yet another exemplary master console handle constructed in accordance with the principles of the present invention.

[0055] FIGS. 11A and 11B show an exemplary slave console constructed in accordance with the principles of the present invention.

[0056] FIG. 12 shows a left slave console constructed in accordance with the principles of the present invention.

[0057] FIG. 13A shows an exemplary controller of the remotely actuated surgical robot system.

[0058] FIG. 13B shows another exemplary controller of the remotely actuated surgical robot system.

[0059] FIGS. 14A-14E show Scara movement of the slave console in accordance with the principles of the present invention.

[0060] FIGS. 15A-15C show vertical adjustment of the slave console in accordance with the

principles of the present invention.

[0061] FIG. **16** shows the slave console in a home configuration in accordance with the principles of the present invention.

[0062] FIGS. **17A-17D** show movement of an exemplary translational instrument interface coupled to the slave console in a forward configuration during zero-degree angulation of the slave console.

[0063] FIGS. **18A-18D** illustrates the forward surgical workspace of FIGS. **17A-17D**.

[0064] FIG. **18E** is a back view of the forward surgical workspace of the slave console of FIGS. **18A-18D**.

[0065] FIGS. **19A-19C** show the forward surgical workspace of an exemplary instrument coupled to the slave console in a forward configuration during twenty-degree angulation of the slave console.

[0066] FIGS. **20A-20C** show the forward surgical workspace of an exemplary instrument coupled to the slave console in a forward configuration during forty-degree angulation of the slave console.

[0067] FIGS. **21A-21J** show flipping of the slave console between a forward configuration and a reverse configuration in accordance with the principles of the present invention.

[0068] FIGS. **21K** and **21L** are a schematic of the master console and slave console during the forward configuration and the reverse configuration, respectively, in accordance with the principles of the present invention.

[0069] FIGS. **22A-22C** show an exemplary translational instrument interface coupled to the slave console in a reverse configuration during zero, twenty, and forty-degree angulation, respectively, of the slave console.

[0070] FIGS. **23A-23C** illustrates the reverse surgical workspaces of FIGS. **22A-22C**.

[0071] FIGS. **24A-24D** show adjustment of the slave console for integrated laparoscopy in accordance with the principles of the present invention.

[0072] FIG. **25** is a flow chart illustrating use of the remotely actuated surgical robot system of FIG. **1** in accordance with the principles of the present invention.

[0073] FIG. **26** is a flow chart illustrating the surgeon console positioning step of FIG. **25** in accordance with the principles of the present invention.

[0074] FIG. **27** is a flow chart illustrating the preparation step of FIG. **25** in accordance with the principles of the present invention.

[0075] FIG. **28** is a flow chart illustrating the ready for instrument step of FIG. **25** in accordance with the principles of the present invention.

[0076] FIG. **29** is a flow chart illustrating the ready for operation step of FIG. **25** in accordance with the principles of the present invention.

[0077] FIG. **30** is a flow chart illustrating the operating step of FIG. **25** in accordance with the principles of the present invention.

[0078] FIGS. **31A** and **31B** show an exemplary remotely actuated surgical robot system having hybrid telemanipulators constructed in accordance with the principles of the present invention.

[0079] FIGS. **32A** and **32B** show partially exploded perspective views of the surgical robot system of FIGS. **31A** and **31B**.

[0080] FIG. **33** shows a top view, partially exploded, of an exemplary mechanical transmission system constructed in accordance with the principles of the present invention.

[0081] FIGS. **34A** and **34B** show side perspective views of an exemplary master unit constructed in accordance with the principles of the present invention.

[0082] FIGS. **34C** and **34D** show alternative embodiments of a handle suitable for use with the master unit depicted in FIGS. **34A** and **34B**.

[0083] FIGS. **35A** and **35B** show side perspective views of an exemplary slave unit constructed in accordance with the principles of the present invention.

[0084] FIGS. **36A** and **36B** show, respectively, an end sectional and side interior perspective view of an exemplary slave hub.

[0085] FIGS. **36C** and **36D** are, respectively, a perspective side view of a slave instrument and a detailed interior view of an end-effector constructed in accordance with the principles of the present invention.

[0086] FIG. **36E** is a detailed view of an alternative embodiment of an exemplary end-effector.

[0087] FIG. **37** shows a flow chart illustrating exemplary method steps for identifying the kinematics of a selected end-effector.

[0088] FIG. **38** shows an alternative exemplary embodiment of a remotely actuated surgical robot system of the present invention.

[0089] FIG. **39** shows interior side perspective view of the master unit of the remotely actuated surgical robot system of FIG. **38**.

[0090] FIGS. **40A** and **40B** are forward and rearward perspective views of the slave unit of the remotely actuated surgical robot system of FIG. **38**.

[0091] FIGS. **40C**, **40D**, and **40E** show another exemplary incision pointer constructed in accordance with the principles of the present invention.

[0092] FIGS. **41A** and **41B** are alternative schematic illustrations of a control system suitable for use in the surgical robot system of the present invention.

[0093] FIGS. **42A** and **42B** are side perspective views of alternative embodiments of telemanipulators constructed in accordance with the principles of the present invention.

[0094] FIG. **43** shows another exemplary master console constructed in accordance with the principles of the present invention.

DETAILED DESCRIPTION

[0095] A remotely actuated surgical robot system having robotic telemanipulators and integrated laparoscopy, which may be used in minimally invasive surgical procedures or in other applications, constructed in accordance with the principles of the present invention, is described herein. The surgical robot system provides the value of robotics for long and difficult surgical tasks such as suturing and dissection, and permits a user, e.g., a surgeon, to efficiently switch to integrated laparoscopy for short and specialized surgical tasks such as vessel sealing and stapling. The fully articulated instruments simplify complex surgical tasks, and replication of hand movements increase precision. The user may be seated or standing in a relaxed ergonomic working position to improve surgeon focus and performance.

[0096] Referring to FIG. **1**, exemplary remotely actuated surgical robot system **10** having robotic telemanipulators is described. Surgical robot system **10** includes master console **20** electrically and operatively coupled to slave console **50** via, e.g., electrical cables. As described in further detail below, surgical robot system **10** includes a macro-synchronization state where a plurality of actuators, e.g., preferably motors, coupled to slave console **50** applies translational macro-movements to an end-effector of slave console **50** responsive to movement applied at master console **20** via a processor-driven control system, and a micro-synchronization state where a plurality of actuators, e.g., preferably motors, coupled to slave console **50** applies micro-movements to an end-effector of slave console **50** responsive to movement applied at a handle of master console **20** via the processor-driven control system.

[0097] The control system may include master controller **2** operatively coupled to right master telemanipulator **22a** and left master telemanipulator **22b** of master console **20**, and slave controllers **4a** and **4b** operatively coupled to right slave telemanipulator **51a** and left slave telemanipulator **51b** of slave console **50**, respectively. For example, master controller **2** may include non-transitory computer readable media, e.g., memory, having instructions stored thereon that, when executed by one or more processors of master controller **2**, allow operation of master console **20**. Similarly, slave controllers **4a** and **4b** may each include non-transitory computer readable media, e.g., memory, having instructions stored thereon that, when executed by one or more processors of respective slave controllers **4a**, **4b**, allow operation of slave console **50**. Master controller **2** is operatively coupled to slave controller **4a** and slave controller **4b** via communication links such as

cables (as illustrated) or via wireless communication components.

[0098] Master controller **2** may be operatively coupled to one or more sensors of master console **20**, and slave controllers **4a**, **4b** may be operatively coupled to one or more actuators of slave console **50** such that master controller **2** may receive signals indicative of movement applied at master console **20** by the one or more sensors of master console **20**, and execute instructions stored thereon to perform coordinate transforms necessary to activate the one or more actuators of slave console **50**, send the processed signals to respective slave controllers **4a**, **4b** that execute instructions stored thereon to move slave console **50** in a manner corresponding to movement of master console **20** based on the processed signals. For example, the one or more actuators may include one or more motors. Alternatively, master controller **2** may receive the signals from the one or more sensors of master console **20**, process the signals, and transmit the processed signals to respective slave controllers **4a**, **4b** which execute instructions stored thereon to perform the coordinate transforms based on the processed signals, and execute instructions to activate the one or more actuators of slave console **50** to move slave console **50** in a manner corresponding to movement of master console **20** based on the transformed, processed signals. Preferably, the slave links and joints of slave console **50** move in a manner such that the end-effector/instrument tip replicates the movement applied at the handle of master console **20**, without deviating, during operation of surgical robot system **10**, from a remote center-of-motion, as described in further detail below. Thus, translation degrees-of-freedom, e.g., left/right, upward/downward, inward/outward, the articulation degrees-of-freedom, e.g., pitch and yaw, the actuation degrees-of-freedom, e.g., open/close, and the rotation degree-of-freedom, e.g., pronosupination, are electromechanically replicated via sensors, actuators, and a control system as described in further detail below.

[0099] In accordance with one aspect of the present invention, the slave links and joints of slave console **50** may move in a manner responsive to movement applied at the handle of master console **20** such that the surgical instrument reproduces the movement applied at the handle of master console **20** at a scaled degree. For example, master controller **2** may receive the signals from the one or more sensors of master console **20**, process the signals, and transmit the processed signals to respective slave controllers **4a**, **4b** which execute instructions stored thereon to perform the coordinate transforms based on the processed signals to generate signals indicative of corresponding scaled movements, and execute instructions to activate the one or more actuators of slave console **50** to move slave console **50** in a scaled manner corresponding to movement of master console **20** based on the transformed, scaled processed signals. Accordingly, the surgeon may apply a micro roll movement of, e.g., 30 degrees, to the handle of master console **20**, such that the actuators of slave console **50** cause the surgical instrument of slave console **50** to perform a micro roll movement at a scaled degree of, e.g., 60 degrees, thereby resulting in a scaled ratio movement (e.g., 1:2) by slave console **50** in response to movement at master console **20**.

Preferably, the micro-scaling causes greater micro movements at the end-effector than occurred at the handle. Each of the macro degrees-of-freedom, e.g., translation, and each of the micro degrees-of-freedom, e.g., articulation, actuation, and rotation, may be independently scaled such that corresponding movement at slave console **50** in an assigned degree-of-freedom is selectively scaled in comparison to the movement applied at master console **20** by the surgeon. For example, the rotation micro degree-of-freedom may be programmed to have corresponding scaled movements between the master and slave at a first scale (1:2), whereas the actuation and/or articulation micro degrees-of-freedom may be programmed to have corresponding scaled movements between the master and slave at a second scale (2:3), different than the first scale. A third scale may be used for the third micro degree-of-freedom. As will be understood by a person having ordinary skill in the art, the scale ratios may be, e.g., 3:1, 2.5:1, 2:1, 1.5:1, 1:1.5, 1:2, 1:2.5, or 1:3, and may vary across the degrees-of-freedom. Advantageously, when micro-scaling calls for greater scaled movements at the end-effector than at the handle, the surgeon need not apply a large movement on the handle of

master console **20** to achieve a desired large movement by the surgical instrument of slave console **50**. This allows for more efficient aspects of robotic surgery (e.g., suturing) with reduced strain on the surgeon's hand/wrist/arm.

[0100] Master console **20** may be positioned within the operating room where a user, e.g., surgeon, may be situated, and in close proximity to slave console **50** where a patient undergoing surgery may be situated, e.g., the sterile zone, so that the user may move quickly between master console **20** and slave console **50** to manually perform laparoscopy during the surgery if necessary.

Accordingly, slave console **50** is designed to efficiently retract to a configuration to permit the surgeon to access the surgical site on the patient as described in further detail below. Master console **20** may be covered with a sterile drape, and may include removable handles that may be removed and sterilizable between surgeries such that the handles are sterile during the surgery and there are no physical barriers between the handles and the surgeon's hands, thereby improving control and performance by the surgeon. The removable handle may be purely mechanical without electronics such as circuits, sensors, or electrically coupled buttons so that the removable handle is easily sterilizable between surgeries. In this manner, the master console may be sterile during the surgery while permitting the surgeon to have the tactile feedback available from direct contact with the robot's handles.

[0101] As illustrated in FIG. **1**, master console **20** includes right master telemanipulator **22a** and left master telemanipulator **22b**. Right master telemanipulator **22a** and left master telemanipulator **22b** may be positioned on a single master console such that right master telemanipulator **22a** may be manipulated by the surgeon's right hand and left master telemanipulator **22b** may be manipulated by the surgeon's left hand when the surgeon is situated at master console **20**.

Accordingly, master console **20** may include wheels for mobility within the operating room, and wheel locks that may be actuated to lock the telemanipulators in position, e.g., during storage or during use by the surgeon during the surgery. In addition, right master telemanipulator **22a** and left master telemanipulator **22b** may be operated simultaneously and independently from the other, e.g., by the surgeon's right and left hands. Preferably, surgical robot system **10** is optimized for use in surgical procedures.

[0102] As further illustrated in FIG. **1**, slave console **50** includes right slave telemanipulator **51a** operatively coupled to right master telemanipulator **22a**, and left slave telemanipulator **51b** operatively coupled to left master telemanipulator **22b**. Right and left slave telemanipulators **51a** and **51b** may be positioned on separate consoles such that right slave telemanipulator **51a** may be positioned on the right side of the patient undergoing surgery and left slave telemanipulator may be positioned on the left side of the patient. Accordingly, right and left slave telemanipulators **51a** and **51b** each may include wheels for mobility within the operating room, and floor locks that may be actuated to lock the telemanipulators in position, e.g., during storage or adjacent the patient during the surgery. In addition, right and left slave telemanipulators **51a** and **51b** each may include a pull bar for pushing and pulling the telemanipulators within the operating room.

[0103] Moreover, a camera system may be used with surgical robot system **10**. For example, a camera e.g., an endoscope, that is manipulated by the assistant situated at slave console **50** may be operated and/or held in position at slave console **50**. Accordingly, the camera system may include display **21** mounted on master console **20** in a position that is easily observable by the surgeon during a surgical procedure. Display **21** may display status information on the surgical robot system **10**, and/or display the surgical site captured by the endoscopic camera to surgeon in real-time.

[0104] Referring now to FIG. **2A**, exemplary master console **20** is described. As described above, master console **20** includes right master telemanipulator **22a** and left master telemanipulator **22b**. As left master telemanipulator **22b** may be a structurally mirrored version of right master telemanipulator **22a** as illustrated, the description below of right master telemanipulator **22a** applies also to left master telemanipulator **22b**.

[0105] Master telemanipulator **22a** includes a plurality of master links, e.g., first master link **26**,

second master link **28**, third master link **30**, and fourth master link, e.g., guided master link **32**, interconnected by a plurality of master joints, e.g., first master joint **25**, second master joint **27**, third master joint **29**, fourth master joint **31**, and fifth master joint **34**. As shown in FIG. **1**, handle portion **35** is connected to master telemanipulator **22a** via joint **34**, and includes a plurality of handles links interconnected by a plurality of handle joints for operating master telemanipulator **22a**. In addition, master telemanipulator **22a** includes a base portion having telescoping bases **23a** and **23b**, and base cap **24** fixed atop telescoping bases **23a** and **23b**. Link **26** is rotatably coupled to base cap **24** via joint **25**. Thus, link **26**, and accordingly all the master joints and links distal to link **26**, may rotate relative to base cap **24** about axis $\delta_{sub.1}$ at joint **25**. As shown in FIG. **1**, link **28**, and accordingly all the master joints and links distal to link **28**, may rotate relative to link **26** about axis $\delta_{sub.2}$ at joint **27**, link **30**, and accordingly all the master joints and links distal to link **30**, may rotate relative to link **28** about axis $\delta_{sub.3}$ at joint **29**, and guide master link **32**, and accordingly all the master joints and links distal to guided master link **32**, may rotate relative to link **30** about axis $\delta_{sub.4}$ at joint **31**.

[0106] Master console **20** includes a plurality of sensors positioned within master telemanipulator **22a** such that any movement applied to any master links and joints may be sensed and transmitted to the control system, which will then execute instructions to cause one or more actuators coupled to slave console **50** to replicate the movement on corresponding slave link and joints of slave telemanipulator **51a**, as described in further detail below with reference to FIG. **12**.

[0107] Still referring to FIG. **2A**, master telemanipulator **22a** includes mechanical constraint **33**, which includes an opening within link **26** sized and shaped to permit guided master link **32** to be positioned therethrough, thereby constraining movements of master telemanipulator **22a** about a pivot point at master telemanipulator **22a**. For example, mechanical constraint **33** ensures that, when master telemanipulator **22a** is actuated, guided master link **32** translates along longitudinal axis **85**. In addition, mechanical constraint **33** enables guided master link **32** to rotate about axes **61** and **86** that are perpendicular to each other, creating a plane that intersects longitudinal axis **85** at stationary pivot point **P** independently of the orientation of guided master link **32**. As a result, the slave telemanipulator produces corresponding movements, thereby virtually maintaining the pivot point of the master telemanipulator, for example, at the fixed incision point on a patient where a trocar passes into a patient's abdomen.

[0108] When surgical robot system **10** is positioned such that remote center-of-motion **Vis** aligned with the patient incision, translational movement applied to handle portion **35** is replicated by the end-effector disposed inside the patient. Because the end-effector replicates the movement applied to handle portion **35**, this arrangement advantageously eliminates the fulcrum effect between the handle and end-effector.

[0109] In addition, master console **20** may include arm support **12**, e.g., coupled to base cap **24**, sized and shaped to permit the surgeon to rest the surgeon's arms against the arm support during operation of master console **20**. Accordingly, arm support **12** remains static during operation of master telemanipulator **22a**. Master console **20** further may include clutch **11**, e.g., a foot pedal, that when actuated prevents macro-synchronization and/or micro-synchronization of surgical robot system **10**, as described in further detail below. As such, master console **20** permits macro-clutching and micro-clutching.

[0110] Referring now to FIG. **2B**, display **21** is described. Display **21** may have a simplistic design without text, utilizing only visible graphical elements and LEDs, e.g., white, yellow, and red lights. For example, white light conveys that the component is functioning properly, yellow light conveys that the surgeon has conducted an inappropriate action, and red conveys that there is an error with the component. As shown in FIG. **2B**, display **21** graphically displays various components of slave console **50** and the status thereof. Icon **21a** corresponds with the booting of the system, icon **21b** corresponds with a system warning, icon **21h** corresponds with the work limit being reached, and icon **21j** corresponds with whether the respective slave telemanipulators of slave console **50** are in

a forward surgical workspace or a reverse surgical workspace, all of which may be invisible when not lit up, whereas all other icons have a visible graphical element even when not lit up. Icon **21c** corresponds with homing, e.g., home configuration, of slave console **50**, icon **21d** corresponds with the status of instrument **82**, icon **21e** corresponds with the sterile interface of translation instrument interface **81**, icon **21f** corresponds with macro-synchronization, icon **21g** corresponds with micro-synchronization, and icon **21i** corresponds with whether the wheels of slave console **50** are locked or unlocked, the functionality of all of which will be described in further detail below. As will be understood by a person having ordinary skill in the art, display **21** may be any display known in the art that may convey information to the surgeon.

[0111] FIG. 2C shows another exemplary master console similar to that shown in FIG. 2A except master console **20** in FIG. 2C further includes an additional clutch **11'**, e.g., an additional foot pedal. In this manner, actuation of clutch **11** allows the master/slave to transition between one type of synchronization/desynchronization (e.g., for macro movements) and actuation of clutch **11'** allows the master/slave to transition between another type of synchronization/desynchronization (e.g., for micro movements, for both micro and macro movements), as described in further detail below. As such, independently actuatable macro-clutching and micro-clutching is permitted. Alternatively or additionally, different predetermined actuation patterns (e.g., multiple presses at the pedal within a predetermined time versus one press at the pedal within the predetermined time) at clutch **11** and/or clutch **11'** may be used for independently actuatable macro-clutching and micro-clutching.

[0112] Referring now to FIGS. 3A-3C, master console **20** may be adjusted between a seated configuration and a standing configuration via telescoping bases **23a** and **23b**. For example, as illustrated in FIG. 3A, master console **20** may be adjusted to a seated configuration such that telescoping bases **23a** and **23b** have a vertical height D.sub.1. In this seated configuration, the surgeon may be seated during operation of master console **20**. As illustrated in FIGS. 3B and 3C, master console **20** may be adjusted to a standing configuration such that telescoping bases **23a** and **23b** have a vertical height D.sub.2. In this configuration, the surgeon may be standing during operation of master console **20**. In addition, the vertical height of telescoping bases **23a** and **23b** may be adjusted via an actuator positioned on master console **20**, e.g., on master link **26**. For example, the actuator may include up and down buttons that when actuated, cause the vertical height of telescoping bases **23a** and **23b** to increase or decrease, respectively. As will be understood by a person having ordinary skill in the art, the vertical height of telescoping bases **23a** and **23b** may be adjusted to any vertical height between D.sub.1 and D.sub.2, as desired by the surgeon.

[0113] Referring now to FIG. 4, master console handle portion **35** is described. Master console handle portion **35** includes a plurality of handle links, e.g., handle link **36** and handle link **38**, interconnected by a plurality of handle joints, e.g., handle joint **37** and handle joint **39**. As illustrated in FIG. 4, handle link **36** is rotatably coupled to guided master link **32** via joint **34**, and accordingly may rotate relative to guided master link **32** about axis **87**. In addition, handle link **38** is rotatably coupled to handle link **36** via handle joint **37**, and accordingly may rotate relative to handle link **36** about axis **88**. Moreover, handle grip **40** may be removably coupled to master console handle portion **35** at joint **39**, such that handle grip **40** may rotate relative to handle link **37** about axis **89**. As shown in FIG. 4, handle grip **40** may include finger strap **41** for engagement with the surgeon's fingers, e.g., thumb and index finger.

[0114] Inward/outward movement of handle portion **35** causes guided master link **32** to move inward/outward along longitudinal axis δ .sub.5, the movement of which is sensed by one or more sensors coupled to master telemanipulator **22a** and transmitted to the control system, which then executes instructions to cause one or more actuators coupled to slave telemanipulator **51a** to cause the corresponding slave link to replicate the inward/outward movement about virtual longitudinal axis ω .sub.9. Similarly, upward/downward movement of handle portion **35** causes guided master link to move upward/downward along longitudinal axis δ .sub.6, the movement of which is sensed

by one or more sensors coupled to master telemanipulator **22a** and transmitted to the control system, which then executes instructions to cause one or more actuators coupled to slave telemanipulator **51a** to cause the corresponding slave link to replicate the upward/downward movement about virtual longitudinal axis $\omega_{sub.10}$. Finally, left/right movement of handle portion **35** causes guided master link to move left/right along longitudinal axis $\delta_{sub.1}$, the movement of which is sensed by one or more sensors coupled to master telemanipulator **22a** and transmitted to the control system, which then executes instructions to cause one or more actuators coupled to slave telemanipulator **51a** to cause the corresponding slave link to replicate the left/right movement about virtual longitudinal axis $\omega_{sub.5}$.

[0115] Still referring to FIG. **4**, movement applied at handle portion **35** of master telemanipulator **22a** actuates the articulation degrees-of-freedom, e.g., pitch and yaw, the actuation degree-of-freedom, e.g., open/close, and the rotation degree-of-freedom, e.g., pronosupination, electromechanically via sensors, actuators, and the control system. Master telemanipulator **22a** preferably includes one or more sensors coupled to handle portion **35** for detecting motion of handle portion **35**. As will be understood, the sensors may be any sensor designed to detect rotational movement, such as magnetic-based rotational sensors that includes a magnet on one side and a sensor on another side to measure rotation by measuring angle and position. The sensors are coupled to a control system for generating signals indicative of the rotation measured by the sensors and transmitting the signals to one or more actuators coupled to slave console **50**, which may reproduce movements applied on handle portion **35** to the end effector. For example, electrical cables may extend from handle portion **35** to the control system, e.g., a unit containing control electronics, and additional electrical cables may extend from the control system to the one or more actuators coupled to slave console **50**.

[0116] As illustrated in FIG. **5A**, handle grip **40** includes triggers **41a**, **41b** that are biased toward an open configuration. Accordingly, triggers **41a**, **41b** may be actuated to generate a signal that is transmitted via the control system, which executes instructions that causes the actuators coupled to slave console **50** to actuate the end-effector to open/close.

[0117] Referring back to FIG. **4**, handle grip **40** may be rotatable about handle axis $\delta_{sub.9}$, such that rotation of the handle grip **40** is detected by a sensor that generates and transmits a signal via the control system, which executes instructions that causes the actuators coupled to slave console **50** to cause rotation of the end-effector in the pronosupination degree-of-freedom.

[0118] Handle portion **35** also is rotatable about handle axis $\delta_{sub.8}$, such that the rotation about handle axis $\delta_{sub.8}$ is detected by a sensor, which generates and transmits a signal via the control system, which executes instructions that causes the actuators coupled to slave console **50** to cause movement of the end-effector in the yaw degree-of-freedom. In addition, handle portion **35** may be rotatable about handle axis **87**, such that the rotation of handle portion **35** about handle axis $\delta_{sub.7}$ is detected by a sensor, which generates and transmits a signal via the control system, which executes instructions that causes the actuators coupled to slave console **50** to cause movement of the end-effector in the pitch degree-of-freedom.

[0119] As illustrated in FIGS. **5B** and **5C**, handle grip **40** may be removably coupled to handle portion **35** of master telemanipulator **22a** via joint **39**. Accordingly, handle grip **40** may be removed between surgeries to be sterilized, and reconnected to master telemanipulator **22a** just before a surgery. Thus, as the entirety of master console **20** may be covered with a sterile drape during operation of surgical robot system **10**, handle grip **40** will be sterile and may be connected to master console **20** outside of the sterile drape. This permits the surgeon to directly contact handle grip **40** without a physical barrier therebetween, thereby improving tactile feedback and overall performance.

[0120] Referring now to FIGS. **5D-5F**, handle grip **40** may be removably coupled to handle portion **35** of master console **20** via a clip attachment. As shown in FIGS. **5D-5F**, spring **43** may be connected to joint **39** of handle portion **35** and clip portion **42** of handle grip **40** to preload the

attachment to eliminate fixation backlash.

[0121] As shown in FIG. 5G sterile drape cap **13** may be removably coupled to master console handle portion **35** in accordance with the principles of the present invention. As described above, master console **20** may be covered with sterile drape **14** during the surgery while permitting the surgeon to have the tactile feedback available from direct contact with the robot's handles. Sterile drape interface **14** includes sterile drape ring **14a** defining opening **14b** within sterile drape interface **14**, and sterile drape **14c** coupled to sterile drape ring **14a**. While the handle grip is not attached to master console **20**, e.g., during sterilization and/or cleaning, sterile drape cap **13** may be temporarily coupled to sterile drape ring **14a** and to master console handle portion **35** to avoid accidental contact with the interior of the master console handle interface by the clinician. For example, sterile drape cap **13** may be fitted over master console handle portion **35** and held in place via methods known in the art including, but not limited to, a system of magnets, friction forces, Velcro surfaces, matching geometries, hooks, etc. When the handle grip is ready to be coupled to master console handle portion **35**, sterile drape cap **13** may be removed and disposed of. Sterile drape ring **14a** is preferably formed of a rigid material, e.g., metal, and is designed to be sandwiched between master console **20** (e.g., at master console handle portion **35**) and the handle grip when the handle grip is coupled to the master console. This provides secure coupling of sterile drape **14c** to the master console using the removable handle. In addition, sterile drape interface **14** may be easily removed from the master console when the handle grip is removed for sterilization for further surgeries. Opening **14b** of sterile drape interface **14** allows components of the handle to move into the master console responsive to actuation by the surgeon through opening **14b** of sterile drape interface **14** without interference of sterile drape **14c**. This ensures a sterile surgery while allowing interaction between the removable handle and the master console.

[0122] In accordance with another aspect of the present invention, as shown in FIG. 6, handle grip **40'** may be removably coupled to handle portion **35'** of master console **20** via a screw attachment. As shown in FIG. 6, screw portion **42'** of handle grip **40'** having inner threaded portion **44a** may engage with outer threaded portion **44b** at joint **39'** of handle portion **35'**, such that handle grip **40'** is screwed onto handle portion **35'**.

[0123] Referring now to FIGS. 7A-7C, actuation steps of handle grip **40** are described. As illustrated in FIGS. 7B and 7C, handle grip **40** includes retractable piston **45** positioned within a central lumen of handle grip **40**. Retractable piston **45** is mechanically coupled to triggers **41a**, **41b** of handle grip **40** via connectors **46a**, **46b**, respectively. As shown in FIG. 7A, when triggers **41a**, **41b** are in a relaxed state, e.g., biased toward an open configuration, retractable piston **45** is completely within the central lumen of handle grip **40**. As shown in FIGS. 7B and 7C, as handle grip **40** is actuated, e.g., triggers **41a**, **41b** are pressed toward each other, connectors **46a**, **46b** cause retractable piston **45** to protrude out of the central lumen of handle grip **40**. The movement of retractable piston **45** beyond the central lumen of handle grip **40** may be sensed by sensors within handle portion **35**.

[0124] For example, as shown in FIGS. 7D and 7E, the portion of master console adjacent to where handle grip **40** is removably coupled to handle portion **35** may include one or more sensors **47** for sensing movement at handle portion **35**. Accordingly, one or more sensors **47** may transmit a signal to the control system indicating movement of retractable piston **45**, and the control system may execute instructions to cause one or more actuators to cause movement by the end-effector. This may serve as a fail-safe because the control system will not instruct the actuators to cause movement by the end-effector without sensors **47** sensing movement of retractable piston **45**. For example, when triggers **41a**, **41b** of handle grip **40** are in a relaxed state, no movement will be sensed due to small incidental movements of triggers **41a**, **41b** until triggers **41a**, **41b** are purposefully actuated by the surgeon. Thus, triggers **41a**, **41b** may have to be actuated at least a pre-specified amount in order for retractable piston **45** to protrude beyond the central lumen of handle grip **40**. In addition, as illustrated in FIGS. 7D and 7E, handle portion **35** may include spring

48 for pushing against retractable piston **45** to bias triggers **41a**, **41b** in an open configuration via connectors **46a**, **46b**.

[0125] Referring now to FIG. **8A-8G**, another exemplary handle grip is provided. Handle grip **71** is constructed to be coupled to master console handle portion **35** at sterile drape interface **14** (e.g., by sandwiching sterile drape ring **14a** between master console handle portion **35** and the movable outer flange to clip the components together). Handle grip **71** includes triggers **72a** and **72b**, which are biased in an open configuration. For example, as shown in FIG. **8B**, when the clinician does not apply any force to triggers **72a**, **72b**, triggers **72a**, **72b** may be biased to, e.g., 150-160 degrees, or preferably 155 degrees, apart. At any time, when the clinician releases triggers **72a**, **72b**, triggers **72a**, **72b** will return to the predetermined open configuration. Triggers **72a**, **72b** may be biased to an open configuration via springs as described above with reference to triggers **41a**, **41b**. Like handle grip **40**, triggers **72a**, **72b** of handle grip **71** may have to be pressed to be a first predetermined angle from one another before the control system executes instructions to cause one or more actuators to cause movement at the surgical instrument. For example, as shown in FIG. **8C**, the actuators will not cause movement by the plurality of slave links and slave joints or the end-effector until triggers **72a**, **72b** are pressed such that triggers **71a**, **72b** are within the first predetermined angle, e.g., 90 degrees apart or less. This may serve as a fail-safe to prevent small incidental movements of triggers **72a**, **72b** from being replicated by the end-effector until triggers **72a**, **72b** are purposefully actuated by the surgeon. For example, triggers **72a**, **72b** must be pressed to within 90 degrees apart (or another preset angle), before the plurality of slave links and slave joints move responsive to movement of at least one of the plurality of master links and master joints. If triggers **72a**, **72b** are released by the surgeon, the slave console will cease movement responsive to movements at the master console. In this manner, a surgeon can readjust the plurality of master links and master joints to a different desirable configuration without causing corresponding movements at any of the slave links or joints. This allows the surgeon to readjust the master console to a more comfortable position without moving the slave console. Once triggers **72a**, **72b** are reengaged to the first predetermined angle, or less, the controller will then cause corresponding movements at the slave console responsive to movement at the handle. Handle grip **71** further includes palm extension **72c** that extends from the triggers in the direction of a surgeon's palm so that the surgeon's palm can contact palm extension **72c** while performing surgery for ergonomic purposes.

[0126] In addition, while the controller will not cause movement at the slave console unless the surgeon engages the handle in a predetermined manner (e.g., triggers **72a**, **72b** are moved toward one another to within the first predetermined angle or less), the controller also will not cause micro-movements at the end-effector unless the surgeon engages the handle in a second predetermined manner. For example, the second predetermined manner may be to move triggers **72a**, **72b** toward one another a second predetermined angle (e.g., 30 degrees apart) or less, where the second predetermined angle is less than the first predetermined angle. In this manner, the slave console will not move responsive to movement at the handle/master console unless a first actuation pattern is sensed at the handle by the controller (e.g., triggers **72a**, **72b** are moved toward one another to within the first predetermined angle or less) and the end effector will not open/close unless a second actuation pattern is sensed at the handle by the controller (e.g., triggers **72a**, **72b** are moved toward one another to within the second predetermined angle or less).

[0127] Referring now to FIGS. **8D** and **8E**, the internal components of handle grip **71** are described. As shown in FIGS. **8D** and **8E**, handle grip **71** has release grip **76**, which is coupled to coaxial retraction tube **74**, such that retraction of release grip **76** causes retraction tube **74** to retract. Retraction tube **74** has a number of apertures **75** corresponding with a number of hooks **73**, such that each of hooks **73** protrudes through each of apertures **75** when release grip **76** and retraction tube **74** are in a relaxed state. For example, handle grip **71** may have one, two, three, or four hooks. Hooks **73** have an angled surface, e.g., angling away from the direction where handle grip **71** is

coupled to the master console, so that as retraction tube **74** is retracted via release grip **76**, the edge of apertures **75** is moved over the angled surface of hooks **73** causing hooks **73** to bend radially inward toward the center axis of retraction tube **74**. When release grip **76** is released, both release grip **76** and retraction tube **74** returns to their relaxed state allowing apertures **75** to realign with hooks **73** such that hooks **73** protrude through apertures **75**.

[0128] Master console handle portion **35** includes angular orientation constraint **80b**, which has a geometry that corresponds with angular orientation constraint **80a** of handle grip **71** such that retraction tube **74** may be inserted within a lumen of angular orientation constraint **80b** at a specific rotational orientation. As shown in FIG. **8E**, angular orientation constraint **80b** includes one or more grooves **77** on its inner surface having a geometry that corresponds with hooks **73** such that when retraction tube is inserted within the lumen of angular orientation constraint **80b** and hooks **73** are protruding through apertures **75**, hooks **73** will engage with groove **77**. There may be a number of grooves corresponding with the number of hooks **73**, or groove **77** may fully extend circumferentially along the inner surface of angular orientation constraint **80b**. Moreover, handle grip **71** may include actuation rod **78** which is sized to fit within the lumen of preload spacer **86** and interact with actuation rod **79** of the master console. For example, actuation rod **79** may extend a predetermine distance within preload spacer **86** to push against actuation rod **78** of handle grip **71** for biasing triggers **72a**, **72b** in the open configuration as described above.

[0129] Preload spacer **86** may be coupled to preload spring **85**, such that preload spacer pushes against handle grip **71** when handle grip **71** is engaged with master console handle portion **35**. Accordingly, when hooks **73** are engaged with grooves **77**, preload spacer **86** pushes against handle grip **71** to keep hooks **73** positioned within grooves **77** and cancel any backlash. Specifically, the trailing edge of hooks **73** is pushed against the trailing edge of grooves **77** to prevent lateral movement of handle grip **71** relative to master console handle portion **35** when handle grip **71** is engaged.

[0130] As shown in FIG. **8F**, release grip **76** may have a textured surface for ease of retraction of release grip **76** by the clinician when coupling and decoupling handle grip **71** with master console handle portion **35**. As shown in FIGS. **8F** and **8G**, release grip **76** engages with sterile drape interface **14** when handle grip **71** is engaged with master console handle portion **35** such that handle grip **71** may still rotate freely. As shown in FIG. **8F**, sterile drape ring **14a** of sterile drape interface **14** having sterile drape **14c** coupled thereto snaps onto master console handle portion **35**, such that sterile drape **14c** maintains sterility of the master console during operation.

[0131] Referring now to FIGS. **9A-9C**, the steps of coupling handle grip **71** to the master console are illustrated. FIG. **9A** shows release grip **76** of handle grip **71** in a relaxed state, with hooks **73** protruding through apertures **75** of retraction tube **74**. FIG. **9B** illustrates retraction of release grip **76**, when the edge of apertures **75** causes hooks **73** to move radially inward such that retraction tube **74** may be inserted within the lumen of angular orientation constraint **80b**. When retraction tube **74** is fully inserted within the lumen of angular orientation constraint **80b**, release grip **76** may be released as shown in FIG. **9C** to permit hooks **73** to protrude through apertures **75** of retraction tube **74** and to engage with grooves **77** in the relaxed state. FIG. **9C** illustrated handle grip **71** coupled to master console handle portion **35**.

[0132] Referring now to FIGS. **9D-F**, the steps of decoupling handle grip **71** from the master console are illustrated. FIG. **9D** shows release grip **76** being retracted such that the edge of apertures **75** causes hooks **73** to move radially inward and disengage with grooves **77**. Once disengaged, handle grip **71** may be removed from the master console as shown in FIG. **9E**. When handle grip **71** is decoupled from the master console, release grip **71** may be released and returned to its relaxed state, with hooks **73** protruding through apertures **75** of retraction tube **74**.

[0133] In accordance with another aspect of the present invention, as illustrated in FIGS. **10A-10C**, handle grip **40''** may be removably coupled to handle portion **35** of master telemanipulator **22a**. For example, handle grip **40''** may have a pistol shape including a handle and trigger **49** for performing

a desired surgical task. As will be understood by a person having ordinary skill in the art, various shaped handle grips may be removably coupled to the master telemanipulator to actuate a desired movement by the end-effector of the slave telemanipulator. Accordingly, the handle grips may have an integrated identifier element, e.g., an RFID tag, such that the control system detects the identifier element and identifies whether the handle grip is authorized for use with surgical robot system **10**.

[0134] Referring now to FIGS. **11A** and **11B**, slave console **50** is described. As shown in FIG. **11A**, slave console **50** includes right slave telemanipulator **51a** and left slave telemanipulator **51b**. As left slave telemanipulator **51b** may be a structurally mirrored version of right slave telemanipulator **51a** as illustrated, the description below of right slave telemanipulator **51a** applies also to left slave telemanipulator **51b**.

[0135] As illustrated in FIG. **12**, slave telemanipulator **51a** includes a plurality of slave links, e.g., first slave link **55**, second slave link **57**, third slave link **59**, fourth slave link, e.g., angulation link **61**, fifth slave link **63**, sixth slave link **65**, seventh slave link **67**, and eighth slave link, e.g., slave hub **69**, interconnected by a plurality of slave joints, e.g., first slave joint, e.g., proximal Scara joint **54**, second slave joint, e.g., median Scara joint **56**, third slave joint, e.g., distal Scara joint **58**, fourth slave joint, e.g., angulation joint **60**, fifth slave joint, e.g., alpha joint **62**, sixth slave joint, e.g., beta joint **64**, seventh slave joint, e.g., gamma joint **66**, and eighth slave joint, e.g., theta joint **68**. As shown in FIG. **12**, translational instrument interface **81** is coupled to slave telemanipulator **51a** via theta joint **68**.

[0136] Translational instrument interface **81** may be constructed as described in U.S. Patent Application Publication No. 2018/0353252 to Chassot, assigned to the assignee of the instant application, the entire contents of which are incorporated by reference herein. For example, translational instrument interface **81** includes slave hub **69** and a surgical instrument. As shown in FIG. **11B**, slave hub **69** may be affixed to link **67** of slave telemanipulator **51a**. The surgical instrument includes the end-effector disposed at the distal end of the shaft of the surgical instrument, and may be coupled to slave hub **69**. For example, the end-effector may be removably coupled to slave hub **69**. A sterile interface may be positioned between slave hub **69** and the surgical instrument. In addition, translational instrument interface **81** includes a translation transmission system that extends from one or more actuators positioned within slave hub **69** to the components of the end-effector. For example, the end-effector includes a plurality of end-effector links interconnected by a plurality of end-effector joints coupled to the translation transmission system of translational instrument interface **81**, such that actuation of the translation transmission system by the one or more actuators causes movement of the end-effector via the plurality of end-effector links and joints.

[0137] In addition, slave telemanipulator **51a** includes base portion **52** having and adjustable column, and slave support **53** fixed atop the adjustable column. Link **55** is rotatably coupled to slave support **53** via proximal Scara joint **54**. Thus, link **55**, and accordingly all the slave joints and links distal to link **55**, may rotate relative to slave support **53** about axis θ_1 at proximal Scara joint **54**. As shown in FIG. **12**, link **57**, and accordingly all the slave joints and links distal to link **57**, may rotate relative to link **55** about axis θ_2 at median Scara joint **56**, link **59**, and accordingly all the slave joints and links distal to link **59**, may rotate relative to link **57** about axis θ_3 at distal Scara joint **58**, angulation link **61**, and accordingly all the slave joints and links distal to angulation link **61**, may rotate relative to link **59** about axis $\omega_{sub.4}$ at angulation joint **60**, link **63**, and accordingly all the slave joints and links distal to link **63**, may rotate relative to angulation link **61** about alpha axis $\omega_{sub.5}$ at alpha joint **62**, link **65**, and accordingly all the slave joints and links distal to link **65**, may rotate relative to link **63** about beta axis θ_6 at beta joint **64**, link **67**, and accordingly all the slave joints and links distal to link **67**, may rotate relative to link **65** about gamma axis θ_7 at gamma joint **66**, and slave hub **69**, and accordingly translational instrument interface **81** when translational instrument interface **81** is coupled to slave hub **69**, may rotate relative to link **67** about theta axis

ω.sub.8 at theta joint **68**.

[0138] The column integrated into slave support **53** contains an actuator, e.g., an electric motor, that allows for extending and retracting the column, thereby adjusting the height of all links distal to slave support **53** relative to the ground. Alternatively, instead of a column integrated into slave support **53**, slave support **53** may include a mechanical linear guidance system having a counter-balance system based on a counter-weight, and an electric brake to block the vertical movement. Accordingly, when the electric brake is released, the vertical height of all links distal to slave support **53** may be adjusted relative to the ground. Proximal Scara joint **54**, median Scara joint **56**, and distal Scara joint **58** each contain an electric brake that may block the movement of the corresponding joint when the respective brake is engaged and permit manual movement of the respective joint when the respective brake is released. Angulation joint **60** contains an actuator, e.g., an electromagnetic motor, that allows for adjustment of the angular position of link **61** about link **59**. Alpha joint **62**, beta joint **64**, gamma joint **66**, and theta joint **68** are each linked to a dedicated electromagnetic motor and brake pair such that the control system may adjust the angular position of each joint by applying a position command to the respective motor, and stop any movement of the joint by activating the respective brake.

[0139] As will be understood by a person having ordinary skill in the art, slave console **50** may include a plurality of sensors and master console **20** may include a plurality of actuators such that movement applied at slave console **50** may cause movement to be applied at master console **20**, thereby providing tactile feedback.

[0140] Referring now to FIG. **13A**, controller **70** is described. Controller **70** may be a remote controller or a graphical user interface operatively coupled to the control system of surgical robot system **10**, or a series of actuators integrated into the left and right telemanipulator **51a** and **51b**, respectively. Accordingly, controller **70** may include a plurality of actuators, e.g., buttons, or a touchscreen interface whereby a user may select a plurality of options via touch. For example, controller **70** may provide a user with the option to select at least one of the following commands: Scara brake engagement and release **70a**, park position configuration **70b**, flipping from forward to reverse gear or from reverse gear to forward gear **70c**, vertical adjustment of slave console **70d**, vertical column brake release **70e**, laparoscopic configuration **70f**, home configuration **70g**, and increase and decrease the forward angulation **70h**. Controller **70** is operatively coupled to one or both slave controllers and/or the master controller. As shown in FIG. **13A**, controller **70** may be integrated into a link of the slave console itself. For example, controller **70** is integrated into third slave link **59'** of left slave telemanipulator **51b** to control certain functionality of that slave telemanipulator responsive to user input and a second controller with the same functionality is integrated into a slave link (e.g., third slave link) to control certain functionality of that slave telemanipulator, e.g., right slave telemanipulator **51a**, responsive to user input.

[0141] For example, when the user actuates Scara brake engagement and release **70a** interface, the controller will cause Scara brake to transition from engaged to released or vice versa. When the user actuates park position configuration **70b** interface, the controller will cause the slave telemanipulator to move into a position suitable for transportation and storage. When the user actuates flipping from forward to reverse gear or from reverse gear to forward gear **70c**, the controller will cause the slave telemanipulator to move between a forward surgical workspace and a reverse surgical workspace. When the user actuates vertical adjustment of slave console **70d**, the controller will cause vertical adjustment of the slave telemanipulator. When the user actuates vertical column brake release **70e**, the controller will cause the vertical column brake to transition from engaged to released or vice versa, to prevent or permit, respectively, vertical adjustment of the slave telemanipulator. When the user actuates laparoscopic configuration **70f**, home configuration **70g**, the controller will cause the slave hub to move away from the patient undergoing surgery so that the surgeon may quickly and safely move from the master console to the surgical site on the patient to manually perform laparoscopic procedures on the patient. When the user actuates

increase and decrease the forward angulation **70h**, the controller will cause adjustment in the forward angulation of the slave telemanipulator. Responsive to user input at controller **70**, the respective slave controller executes instructions stored thereon to execute the command(s) explained below inputted by the user. Each slave console may include its own dedicated controller **70** or a common controller **70** may be used for both slave consoles.

[0142] Referring now to FIG. **13B**, controller **70'** is described. Controller **70'** is constructed similar to controller **70**, except that controller **70'** is a remote controller separate from the slave console and powered by a wired or wireless connection. Accordingly, controller **70'** may be removed and operated at a distance away from the slave console for convenience of the clinician and/or operator.

[0143] For example, as illustrated in FIGS. **14A-14E**, controller **70** may permit a user to release the brakes in proximal Scara joint **54**, median Scara joint **56**, and distal Scara joint **56** so that the surgeon may manually reposition the slave arm horizontally by grabbing, holding and pushing/pulling the slave arm links distal to proximal Scara joint **54** while slave support **53** of the slave telemanipulator remains stationary. Specifically, during Scara movement, slave links **55**, **57**, **59** are permitted to move about axes $\omega_{sub.1}$, $\omega_{sub.2}$, $\omega_{sub.3}$, at joints **54**, **56**, and **58**, while slave support **53** of the slave telemanipulator remains stationary, and while the slave joints and link distal to slave link **59** are fixed relative to slave link **59**. Accordingly, the user may adjust the distal end of the slave telemanipulator, e.g., slave hub **69**, to a desired position over the patient undergoing surgery.

[0144] As illustrated in FIGS. **15A-15C**, controller **70** may permit a user to select a vertical adjustment of slave console command whereby the control system will execute instructions to cause the actuator, e.g., motor, coupled to the column in slave support **53** to extend or retract. Specifically, during vertical adjustment of the slave telemanipulator, the relative distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator may be adjusted. For example, as shown in FIG. **15A**, the vertical distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator is $H_{sub.1}$, as shown in FIG. **15B**, the vertical distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator is $H_{sub.2}$, and as shown in FIG. **15C**, the vertical distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator is $H_{sub.3}$. Accordingly, the user may adjust the relative distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator to a desired height over the patient undergoing surgery. In the embodiment where the slave console includes a mechanical linear guidance system having a counter-balance system based on a counter-weight, controller **70** may permit a user to select a vertical adjustment of slave console command whereby the control system will execute instructions to cause an electric brake in the column to be released so that the mechanically counter-balanced linear guidance system may move up or down, thereby adjusting the relative distance between slave link **55** and the top surface of base portion **52** of the slave telemanipulator to a desired height over the patient undergoing surgery.

[0145] As illustrated in FIG. **16**, controller **70** may permit a user to select a home configuration command whereby the control system will execute instructions to cause the actuators coupled to beta joint **64**, gamma joint **66**, and theta joint **68** to move the slave links and joints to a retracted position such that slave hub **69** of the slave telemanipulator is in a desirable position for positioning the shaft of translational instrument interface **81** within a trocar within the patient undergoing surgery. In the home position, slave hub **69** will be positioned relative to the trocar within the patient such that an instrument **82** may be inserted through and coupled to slave hub **69**, such that instrument tip **84** will slide into, but not pass through the trocar, hence permitting the surgeon to insert the instrument safely and without the need for supervising the distal end of the trocar with the help of an endoscope.

[0146] In addition, controller **70** may permit a user to select an angulation command whereby the control system will execute instructions to cause the actuator coupled to angulation joint **60** to

adjust the angulation of angulation link **61** about axis ω .sub.4 at angulation joint **60** to a desired angulation angle, e.g., between zero and 45-degrees relative to base **52** of slave telemanipulator **51a**. Specifically, when the angulation command is actuated, angulation link **61**, and accordingly all the slave links and joints distal to angulation link **61**, will rotated about axis ω .sub.4 at angulation joint **60**, while slave link **59** and all the slave links and joints proximal to slave link **59**, and base portion **52** of the slave telemanipulator remains stationary. By adjusting the angulation angle of the slave telemanipulator, the angle of the surgical workspace of the slave telemanipulator will be adjusted, providing more access by the surgeon to the patient via translational instrument interface **81**.

[0147] For example, FIGS. **17A-17D** illustrate movement of translational instrument interface **81** coupled to slave telemanipulator **51a** during zero-degree angulation of the slave console. As shown in FIGS. **17A-17D**, angulation link **61**, and accordingly angulation axis θ 05, are parallel with the longitudinal axis of base **52** of slave telemanipulator **51a**, and perpendicular with the ground floor. During operation of slave telemanipulator **51a**, the control system only executes instructions to cause the actuators coupled to slave console **20** to apply movement to the slave links and joints distal to angulation link **61**. Accordingly, as shown in FIGS. **18A-18D**, translational instrument interface **81** of slave telemanipulator **51a** has forward surgical workspace FSW, e.g., the extent to which translational instrument interface **81** can reach in a forward configuration during zero-degree angulation of the slave console. FIG. **18E** is a back view of forward surgical workspace FSW of the slave console of FIGS. **18A-18D**.

[0148] FIGS. **19A-19C** illustrate movement of translational instrument interface **81** coupled to slave telemanipulator **51a** during 20-degree angulation of the slave console. As shown in FIGS. **19A**, angulation link **61**, and accordingly angulation axis ω .sub.5, is adjusted to a 20-degree angle relative to the longitudinal axis of base **52** of slave telemanipulator **51a**. During operation of slave telemanipulator **51a**, the control system only executes instructions to cause the actuators coupled to slave console **20** to apply movement to the slave links and joints distal to angulation link **61**. Accordingly, as shown in FIGS. **19B**, translational instrument interface **81** of slave telemanipulator **51a** has forward surgical workspace FSW, e.g., the extent to which translational instrument interface **81** can reach in a forward configuration during 20-degree angulation of the slave console. FIG. **19C** is a back view of forward surgical workspace FSW of the slave console of FIG. **19B** during 20-degree angulation of the slave console.

[0149] FIGS. **20A-20C** illustrate movement of translational instrument interface **81** coupled to slave telemanipulator **51a** during 40-degree angulation of the slave console. As shown in FIGS. **20A**, angulation link **61**, and accordingly angulation axis θ s, is adjusted to a 40-degree angle relative to the longitudinal axis of base **52** of slave telemanipulator **51a**. During operation of slave telemanipulator **51a**, the control system only executes instructions to cause the actuators coupled to slave console **20** to apply movement to the slave links and joints distal to angulation link **61**. Accordingly, as shown in FIGS. **20B**, translational instrument interface **81** of slave telemanipulator **51a** has forward surgical workspace FSW, e.g., the extent to which translational instrument interface **81** can reach in a forward configuration during 40-degree angulation of the slave console. FIG. **20C** is a back view of forward surgical workspace FSW of the slave console of FIG. **19B** during 40-degree angulation of the slave console.

[0150] As illustrated in FIGS. **21A-21J**, controller **70** may permit a user to select a flipping command whereby the control system will execute instructions to cause the plurality of actuators coupled to the slave console to move slave telemanipulator **51a** between a forward surgical workspace and a reverse surgical workspace. For example, the control system may cause the plurality of actuators coupled to the slave console to invert slave telemanipulator **51a** from a forward surgical workspace to a reverse surgical workspace, and vice versa. Specifically, during actuation of the flipping command, link **65**, and accordingly all slave links and slave joints distal to link **65**, are rotated about beta joint **64** of slave telemanipulator **51a**. In addition, as link **65** rotates

about beta joint **64**, link **67** rotates relative to link **65** at gamma joint **66**, and slave hub **69** rotates relative to link **67** about theta joint **68**, until slave telemanipulator **51a** is in a reverse surgical workspace configuration. As illustrated in FIGS. **22B-22H**, translational instrument interface **81** is removed from slave hub **69** prior to actuation of the flipping command to prevent translational instrument interface **81** from injury the patient. As slave telemanipulator **51a** is able to flip between a forward surgical workspace and a reverse surgical workspace by simply removing translational instrument interface **81** and actuating the flipping command without having to unlock slave telemanipulator **51a** and move it about the operating room, and without having to actuate the Scara brake release command or the vertical adjustment of slave console command, the user will save a lot of time and be able to quickly continue operating on the patient in a different surgical workspace.

[0151] Referring now to FIGS. **21K** and **21L**, a schematic of the master console and slave console having a forward surgical workspace and a reverse surgical workspace, respectively, is provided. As shown in FIG. **21K**, when the telemanipulators of slave console **50** have a forward surgical workspace, master controller **2** of master console **20** is programmed such that right master telemanipulator **22a** communicates with right slave telemanipulator **51a**, and left master telemanipulator **22b** communicates with left slave telemanipulator **51b**. Accordingly, master controller **2** may receive signals indicative of movement applied at right master telemanipulator **22a** by the one or more sensors of master console **20**, and execute instructions stored thereon to perform coordinate transforms necessary to activate the one or more actuators of slave console **50**, send the processed signals to respective slave controllers **4a** that execute instructions stored thereon to move right slave telemanipulator **51a** in a manner corresponding to movement of right master telemanipulator **22a** based on the processed signals. Similarly, master controller **2** may receive signals indicative of movement applied at left master telemanipulator **22b** by the one or more sensors of master console **20**, and execute instructions stored thereon to perform coordinate transforms necessary to activate the one or more actuators of slave console **50**, send the processed signals to respective slave controllers **4b** that execute instructions stored thereon to move left slave telemanipulator **51b** in a manner corresponding to movement of left master telemanipulator **22b** based on the processed signals.

[0152] As shown in FIG. **21L**, when the telemanipulators of slave console **50** have a reverse surgical workspace, master controller **2** of master console **20** behaves as a switch board and is programmed such that right master telemanipulator **22a** communicates with left slave telemanipulator **51b**, and left master telemanipulator **22b** communicates with right slave telemanipulator **51a**. This is necessary so that the surgeon positioned at master console **20** and viewing the surgery site via display **21** may operate what appears to the surgeon as the “right” slave telemanipulator (left slave telemanipulator **51b** in the reverse surgical workspace) with right master telemanipulator **22a** and what appears to the surgeon as the “left” slave telemanipulator (right slave telemanipulator **51a** in the reverse surgical workspace) with left master telemanipulator **22a**. Accordingly, master controller **2** may receive signals indicative of movement applied at right master telemanipulator **22a** by the one or more sensors of master console **20**, and execute instructions stored thereon to perform coordinate transforms necessary to activate the one or more actuators of slave console **50**, send the processed signals to respective slave controllers **4b** that execute instructions stored thereon to move left slave telemanipulator **51b** in a manner corresponding to movement of right master telemanipulator **22a** based on the processed signals. Similarly, master controller **2** may receive signals indicative of movement applied at left master telemanipulator **22b** by the one or more sensors of master console **20**, and execute instructions stored thereon to perform coordinate transforms necessary to activate the one or more actuators of slave console **50**, send the processed signals to respective slave controllers **4a** that execute instructions stored thereon to move right slave telemanipulator **51a** in a manner corresponding to movement of left master telemanipulator **22b** based on the processed signals.

[0153] Thus, in the forward surgical workspace configuration, master controller **2** communicates with right slave controller **4a** to cause right slave telemanipulator **51a** to move responsive to movement at right master telemanipulator **22a** and master controller **2** communicates with left slave controller **4b** to cause left slave telemanipulator **51b** to move responsive to movement at left master telemanipulator **22b**. Additionally, in the reverse surgical workspace configuration, master controller **2** communicates with left slave controller **4b** to cause left slave telemanipulator **51b** to move responsive to movement at right master telemanipulator **22a** and master controller **2** communicates with right slave controller **4a** to cause right slave telemanipulator **51a** to move responsive to movement at left master telemanipulator **22b**.

[0154] FIG. **22A** illustrates slave telemanipulator **51a** in a reverse configuration during zero-degree angulation of the slave console, FIG. **22B** illustrates slave telemanipulator **51a** in a reverse configuration during 20-degree angulation of the slave console, and FIG. **22C** illustrates slave telemanipulator **51a** in a reverse configuration during 40-degree angulation of the slave console. In addition, as shown in FIGS. **23A-23C**, translational instrument interface **81** of slave telemanipulator **51a** has reverse surgical workspace RSW, e.g., the extent to which translational instrument interface **81** can reach in a reverse configuration during zero-degree angulation, 20-degree angulation, and 40-degree angulation, respectively, of the slave console.

[0155] As illustrated in FIGS. **24A-24D**, controller **70** may permit a user to select a laparoscopic configuration command whereby the control system will execute instructions to cause the plurality of actuators coupled to the slave console to move slave hub **69** from a surgical mode where a plurality of slave links move responsive to movement applied at handle of master console to thereby move the surgical instrument to perform the robotic surgery, to a laparoscopic mode where slave hub **69** is positioned away from the patient undergoing surgery so that the surgeon may quickly and safely move from master console **20** to the surgical site on the patient to manually perform laparoscopic tasks on the patient. Accordingly, in the laparoscopic mode, the plurality of slave links proximal to slave hub **69** are retracted away from the patient while base **52** of slave console **50** remains stationary to expose the surgical site to permit a surgeon to perform non-robotic surgery at the surgical site without interference from the plurality of slave links and slave hub **69**. Specifically, actuation of the laparoscopic configuration command causes link **63**, and accordingly all the slave links and joints distal to link **63**, to rotate about alpha axis $\omega_{sub.5}$ at joint **62** while angulation link **61**, and accordingly all the slave links and joints proximal to angulation link **61** including base **52** of slave telemanipulator **51a** remain stationary, until slave hub **69** is facing away from the patient as shown in FIG. **24D**.

[0156] Accordingly, in a preferred embodiment, the longitudinal axis of at least one link of the slave console (e.g., angulation link **61** and/or link **63**) remains aligned with the remote center-of-motion in both the surgical and laparoscopic modes to allow for seamless transition back-and-forth between the modes. For example, alpha axis $\omega_{sub.5}$ may remain aligned with the remote center-of-motion of slave console **50** during the transition from the surgical mode to the laparoscopic mode. Advantageously, this allows the surgeon to move slave console **50** between the surgical mode and the laparoscopic mode without having to realign angulation link **61** and alpha axis $\omega_{sub.5}$ with the remote center-of-motion of slave console **50**, and accordingly, the incision point on the patient's body, when transitioning back to the surgical mode. In accordance with another aspect of the present invention, upon actuation of the laparoscopic configuration command, the distal slave links of slave console **50** may be retracted away from the patient about an axis other than alpha axis $\omega_{sub.5}$ while base **52** of slave console **50** remains stationary to expose the surgical site, such that alpha axis $\omega_{sub.5}$ does not remain aligned with the remote center-of-motion of slave console **50** during the transition from the surgical mode to the laparoscopic mode. For example, the distal slave links of slave console **50** may rotate about, e.g., axis $\omega_{sub.4}$, axis $\omega_{sub.3}$, axis $\omega_{sub.2}$, or axis $\omega_{sub.1}$, while base **52** of slave console **50** remains stationary to expose the surgical site.

[0157] In addition, the control system will execute instructions to determine whether translational

instrument interface **81** has been decoupled from slave hub **69** of slave console **50**, and accordingly from the patient at the surgical site, such that the laparoscopic configuration command cannot be actuated unless the control system determines that translational instrument interface **81** is not coupled to slave hub **69**. Accordingly, translational instrument interface **81** must be removed from slave hub **69** prior to actuation of the laparoscopic configuration command by the user to transition the slave console from the surgical mode to the laparoscopic mode.

[0158] Referring now to FIGS. **25-30**, exemplary method **90** for using of surgical robot system **10** via the control system is described. As will be understood by one skilled in the art, the steps of the methods described herein may be executed by one or more processors of the control system (e.g., at the master controller, the first slave controller, and/or the second slave controller) that execute instructions stored in one or more memory components responsive to user input. As shown in FIG. **25**, at step **91**, system **10** is powered on. At step **92**, slave console **50** is prepared to be ready for operating on the patient undergoing surgery as further illustrated in FIG. **27**, and at step **93**, master console **20** is positioned to the surgeon's desired configuration during operation as further illustrated in FIG. **26**.

[0159] For example, FIG. **26** illustrates step **93** of positioning master console **20** to the surgeon's desired configuration. Master console **20** may be moved about the operating room via the wheels at its base while the wheels are unlocked. Upon reaching the desired location within the operating room, the wheel locks are activated to keep master console **20** in place. As shown in FIG. **26**, at step **93A**, the master telemanipulators are stationary and telescoping bases **23a**, **23b** have an initial height. A controller operatively coupled to master console **20**, e.g., a button, may then be actuated to adjust the height of telescoping bases **23a**, **23b**, e.g., to increase or decrease the height of telescoping bases **23a**, **23b**, until master console **20** is at the surgeon's desired height at step **93B**. For example, master console **20** may be adjusted to a seated configuration where the surgeon may be seated during operation of master console **20**, or a standing configuration where the surgeon may be standing during operation of master console **20**. Accordingly, the controller may be actuated to return master console **20** to the initial height, e.g., for storage purposes.

[0160] Referring now to FIG. **27**, step **92** of preparation of slave console **50** is described. As shown in FIG. **27**, at step **92A**, the wheel locks of the slave telemanipulator are disengaged such that slave console **50** may be moved about the operating room to the desired location relative to the patient. The wheel locks can only be disengaged when no instrument **82** is inserted into slave hub **69** so as to avoid injuring the patient. As multiple slave telemanipulators may be used, each slave telemanipulator is positioned during step **92A**. When slave console **50** is in the desired position within the operating room adjacent the patient undergoing surgery, the wheel locks of slave console **50** are activated at step **92B** such that slave console **50** is prevented from further movement about the operating room via its wheels. Accordingly, the wheel locks may be disengaged again at step **92A** if slave console **50** needs to be moved to a different desired position.

[0161] At step **92C**, the Scara brake release command has not been actuated and Scara brakes of slave console **50** are on. At step **92D**, the Scara brake release command may be actuated by the user to position the distal end of the slave telemanipulator, e.g., the slave links distal to link **59**, at a desired position over the patient undergoing surgery. Specifically, upon actuation of the Scara brake release command, slave links **55**, **57**, **59** are permitted to move about axes $\omega_{sub.1}$, $\omega_{sub.2}$, $\omega_{sub.3}$, at joints **54**, **56**, and **58**, while slave support **53** of the slave telemanipulator remains stationary, and while the slave joints and link distal to slave link **59** are fixed relative to slave link **59**. When the distal end of the slave telemanipulator is in the desired position over the patient, actuation of the Scara brake release command ceases at step **92C**. In addition, as described above with reference to FIGS. **15A-15C**, the vertical height of the slave telemanipulators may be adjusted such that the distal end of the slave telemanipulator is at a desired height relative to the trocar within the patient. The Scara brake release command can only be enabled when no instrument **82** is present in slave hub **69** so as to avoid injuring the patient.

[0162] Referring again to FIG. 27, at step 92E, angulation link 61 of the slave telemanipulator is stationary relative to slave link 59. For example, the slave telemanipulator may initially have an angulation angle of zero degrees. At step 92F, the angulation command may be actuated to adjust the angulation of angulation link 61 about axis $\omega_{\text{sub.4}}$ at angulation joint 60 to a desired angulation angle, e.g., between zero and 45-degrees relative to base 52 of slave telemanipulator 51a. Specifically, upon actuation of the angulation command, angulation link 61, and accordingly all the slave links and joints distal to angulation link 61, will rotate about axis $\omega_{\text{sub.4}}$ at angulation joint 60, while slave link 59 and all the slave links and joints proximal to slave link 59, and base portion 52 of the slave telemanipulator remains stationary. When the desired angle of angulation of the slave telemanipulator is achieved, actuation of the angulation command ceases at step 92E such that angulation link 61 of the slave telemanipulator is stationary relative to slave link 59. The angulation command may have two buttons, one to increase and another one to decrease the angulation.

[0163] At step 92G, the slave telemanipulator has a forward surgical workspace, or alternatively, at step 92H, the slave telemanipulator has a reverse surgical workspace. During both steps 92G and 92H, instrument 82 cannot be in slave hub 69. If the slave telemanipulator has a forward surgical workspace at step 92G, and the user desires a reverse surgical workspace, the flipping command may be actuated to invert slave telemanipulator 51a from a forward surgical workspace to a reverse surgical workspace. Specifically, upon actuation of the flipping command, link 65, and accordingly all slave links and slave joints distal to link 65, are rotated about beta joint 64 of slave telemanipulator 51a. In addition, as link 65 rotates about beta joint 64, link 67 rotates relative to link 65 at gamma joint 66, and slave hub 69 rotates relative to link 67 about theta joint 68, until slave telemanipulator 51a is in a reverse surgical workspace configuration. Similarly, if the slave telemanipulator has a reverse surgical workspace at step 92H, and the user desires a forward surgical workspace, the flipping command may be actuated to invert slave telemanipulator 51a from a forward surgical workspace to a reverse surgical workspace.

[0164] At step 92I, translational instrument interface 81 is not coupled to slave hub 69 of the slave telemanipulator. At step 92J, a temporary incision pointer may be removably coupled to the slave telemanipulator. For example, the temporary incision pointer is removably coupled to the slave telemanipulator such that it points to virtual remote center-of-motion V located at a predetermined point on axis @s, such that virtual remote center-of-motion V may be brought in coincidence with the surgical incision point, reducing trauma to the patient and improving cosmetic outcomes of the surgery. The temporary incision pointer may be removed prior to installation of the translational instrument interface 81 if necessary. During preparation step 92, instrument 82 should not be coupled to slave hub 69 of the slave telemanipulator. Thus, if instrument 82 is coupled to slave hub 69 of the slave telemanipulator, at step 92K, the control system will prevent further actions until translational instrument interface 81 is removed.

[0165] At step 92L, the slave links and joints distal to link 61 of the slave telemanipulator may be in any position. Accordingly, at step 92M, the home configuration command may be actuated to move the slave links and joints to a retracted position such that slave hub 69 of the slave telemanipulator is in a desirable position for positioning instrument tip 84 within a trocar within the patient undergoing surgery. At step 92N, the slave telemanipulator is in the home position, wherein slave hub 69 is positioned relative to the trocar within the patient such that instrument 82 may be inserted through and coupled to slave hub 69, and the instrument tip 84 will slide into, but not pass through the trocar.

[0166] At step 92O, the laparoscopic configuration command may be actuated to move slave hub 69 away from the patient undergoing surgery so that the surgeon may quickly and safely move from master console 20 to the surgical site on the patient to manually perform laparoscopic tasks on the patient. Specifically, upon actuation of the laparoscopic configuration command, link 63, and accordingly all the slave links and joints distal to link 63, to rotate about alpha axis $\omega_{\text{sub.5}}$ at joint

62 while angulation link **61**, and accordingly all the slave links and joints proximal to angulation link **61** including base **52** of slave telemanipulator **51a** remain stationary, until slave hub **69** is facing away from the patient. At step **92P**, slave hub **69** is in the retracted position.

[0167] At step **92Q**, the sterile interface of translational instrument interface **81** is not coupled to slave hub **69** of the slave telemanipulator. At step **92R**, the sterile interface is coupled to the slave hub, and the control system determines whether the sterile interface is identified, e.g., by reading an RFID tag integrated into the sterile interface. If the sterile interface is not identified, at step **92S**, the control system awaits removal of the sterile interface until the sterile interface is decoupled from slave hub **69** at step **92Q**. If the sterile interface is identified, the sterile interface is successfully installed at step **92T**.

[0168] At step **92U**, the park position command may be actuated to move slave telemanipulator **51b** into a position suitable for transportation and storage. Specifically, upon actuation of the park position command, the vertical column in slave support **53** retracts to a minimum height, the Scara brakes release to fold the Scara arm into a folded position, the angulation returns to zero-degree angulation, and the joints distal to joint **62** move to fold the slave arm into a compact position. Surgical robot system **10** may be powered off if necessary after step **92**.

[0169] If surgical robot system **10** is not powered off after step **92**, at step **94**, the control system determines whether the sterile interface has been successfully installed and whether the floor lock is activated. If it is determined that either the sterile interface has not been successfully installed or that the floor lock is disengaged, surgical robot system **10** must return to preparation step **92** to rectify the above. If it is determined at step **94** that the sterile interface has been successfully installed and that the floor lock is activated, surgical robot system **10** may proceed to step **95**.

[0170] At step **95**, surgical robot system is ready for instrument **82** as shown in FIG. **28**. For example, at step **95A**, the control system of slave console **50** waits for instrument **82** until instrument **82** is coupled to slave hub **69** of the slave telemanipulator. Accordingly, an instrument **82** is selected and inserted within slave hub **69**. To ensure that the instrument doesn't fall out of the slave hub, the user may mechanically lock the instrument into slave hub **69** by rotating the proximal end of the instrument. Slave hub **69** has an integrated sensor to detect whether the instrument is locked. At step **95B**, sensors positioned within slave hub **69** read out an identifier element integrated with the selected instrument, e.g., an RFID tag, where the RFID tag contains identification information of the selected instrument. At step **95C**, the control system determines whether the selected instrument is authorized based on the detection of the RFID tag. If the selected instrument is not authorized, the control system waits until it is removed at step **95D**. When the unauthorized instrument is removed, step **95D** returns to step **95A**. If the selected instrument is authorized and locked within slave hub **69** of the slave telemanipulator at step **95E**, method **90** may proceed to step **96**. If at any time during step **95** the sterile interface is removed, the floor lock is disengaged, the flipping command is actuated, the Scara brake release command is actuated, the home configuration command is actuated, or the incision pointer is inserted, method **90** may return to preparation step **92**.

[0171] At step **96**, surgical robot system **10** is ready for operation. As shown in FIG. **29**, at step **96A**, the control system verifies that instrument **82** is coupled to slave hub **69** of the slave telemanipulator. At step **96B**, the control system detects when the surgeon grabs handle grip **40** of handle portion **35**. As shown in FIGS. **9A** and **9B**, sensors within the handle may detect that the surgeon has grabbed the handle. At step **96C**, clutch **11** is actuated to prepare the control system for macro-synchronization as described in step **97A**.

[0172] As shown in FIG. **30**, surgical robot system **10** may now be operated. For example, at step **97A**, surgical robot system **10** is in a macro-synchronization state, but not in a micro-synchronization state. In the macro-synchronization state, translational macro-movements applied at master console **20** will be sensed and transmitted to the control system, which instructs the actuators coupled to slave console **50** to cause the corresponding slave links and joints to move in a

manner so that the macro-movements (i.e. up/down, left/right, in/out) of instrument tip **84** corresponds to the macro-movements of the handle at the master console **20**. However, in the unsynchronized macro state, the control system does not cause macro-movements applied at master console **20** to be made in a corresponding manner at slave console **50**. For example, in the unsynchronized macro state, macro-movements at master console **20**, whether intentional or unintentional, will permit the master links of master console **20** to move, but will not result in any corresponding movement at slave console **50**.

[0173] In the micro-synchronization state, micro-movements applied at handle portion **35** of master console **20** will be sensed and transmitted to the control system, which instructs the actuators coupled to slave console **50** to cause the instrument tip **84** move in a manner corresponding to those micro-movements applied at handle portion **35** of master console **20**. However, in the unsynchronized micro state, the control system does not cause micro-movements applied at master console **20**/handle portion **35** to be made in a corresponding manner at slave console **50**/end-effector. Thus, at step **97A**, translational macro-movements are replicated, but micro-movements are unsynchronized. Clutch **11** may be actuated to transition surgical robot system **10** to unsynchronized macro state at step **97B** where translation macro-movements may be prevented by master console **20**, and thus not replicated by slave console **50**. For example, clutch **11** may be a foot pedal that when stepped on, maintains surgical robot system **10** in the unsynchronized macro state. Upon release of clutch **11**, surgical robot system **10** returns to the macro-synchronization state at step **97A**. Accordingly, for example, the surgeon may apply a macro-movement to handle portion **35** in the synchronized macro state, e.g., move handle portion **35** inward/outward thereby causing inward/outward movement on the surgical instrument, then actuate clutch **11** to transition the surgical system from the synchronized macro state to the unsynchronized macro state, move handle portion back to its original or another more comfortable position, which will not result in macro-movement of the surgical instrument, release clutch **11** to transition the surgical system from the unsynchronized macro state to the synchronized macro state, and proceed with applying additional macro-movements to handle portion **35**, which will result in corresponding macro-movement at the surgical instrument. Advantageously, this will permit the surgeon to readjust master console **20** for comfort, while slave console **50** remains in a desired position. In a further example, actuation of clutch **11** transitions the surgical system between the synchronized and unsynchronized states for both macro movements and micro movements.

[0174] In addition, the control system may be programmed to detect an actuation pattern by handle portion **35** such that micro-movements at handle portion **35** are not replicated by the end-effector unless the control system detects the actuation pattern. For example, the actuation pattern may include a quick, double actuation of handle grip **40**. Thus, when the user presses handle grip **40** twice repeatedly at step **97C**, the control system detects the actuation pattern, and surgical robot system **10** is in a micro-synchronization state where micro-movements at handle portion **35** will be replicated by the end-effector. When transitioning from an unsynchronized micro state to a micro-synchronization state, the control system executes instructions to cause the micro-position of instrument tip **84** to have the same spatial orientation relative to the instrument shaft **82** as the spatial orientation of handle portion **35** relative to corresponding link **32** of master telemanipulator **22a**. At step **97D**, surgical robot system **10** is fully in both a macro-synchronization state and a micro-synchronization state, e.g., when the end-effector is in the target position for the operation, and the surgeon can use surgical robot system **10** to perform surgical tasks. Upon actuating clutch **11**, at step **97E**, surgical robot system **10** is in a micro-synchronization state, but in the unsynchronized macro state.

[0175] In accordance with another aspect of the present invention, as shown in FIG. **2C**, master console **20** may have second clutch **11'** for use in conjunction with clutch **11** to actuate the macro-synchronization state and/or the micro-synchronization state of surgical robot system **10**. For example, actuation of clutch **11** may cause surgical robot system **10** to transition between a

synchronized and unsynchronized macro state, and actuation of clutch **11'** may cause surgical robot system **10** to transition between a synchronized and unsynchronized micro state. Accordingly, for example, the surgeon may apply a micro-movement to handle portion **35** in the synchronized macro state, e.g., roll handle portion **35** thereby causing roll movement on the surgical instrument, then actuate clutch **11'** to transition the surgical system from the synchronized micro state to the unsynchronized micro state, move handle portion back to its original or another more comfortable position, which will not result in micro-movement of the surgical instrument though may result in macro-movement of the surgical instrument if the surgical system is in a synchronized macro state, release clutch **11'** to transition the surgical system from the unsynchronized micro state to the synchronized micro state, and proceed with applying additional micro-movements to handle portion **35**, which will result in corresponding micro-movement at the surgical instrument. Advantageously, this will permit the surgeon to readjust master console **20** for comfort, while slave console **50** remains in a desired position.

[0176] The surgeon may selectively choose any combination of synchronized and unsynchronized macro and micro states of surgical robot system **10**. In accordance with yet another aspect of the present invention, actuation of clutch **11** may cause surgical robot system **10** to transition between both unsynchronized macro and micro states and synchronized macro and micro states; whereas, actuation of clutch **11'** only causes surgical robot system **10** to transition between an unsynchronized and synchronized micro state. Alternatively, actuation of clutch **11** may cause surgical robot system **10** to transition between both unsynchronized macro and micro states and synchronized macro and micro states; whereas, actuation of clutch **11'** only causes surgical robot system **10** to transition between an unsynchronized and synchronized macro state.

[0177] In accordance with another aspect of the present invention, a remotely actuated surgical robot system having hybrid telemanipulators, which may be used in minimally invasive surgical procedures or in other applications, constructed in accordance with the principles of the present invention, is described herein.

[0178] Referring to FIGS. **31A** and **31B**, exemplary remotely actuated surgical robot system **100** having hybrid telemanipulators is described. Surgical robot system **100** illustratively is affixed atop moveable cart **101**, to which the hybrid telemanipulators also may be mounted for mobility and ease of transport within an operating room. Surgical robot system **100** includes master region **400**, where a surgeon may be situated to operate system **100**, and remote slave region **500** in proximity to the sterile zone, where a patient to be operated upon may be positioned. As shown in FIG. **31B**, the operating surgeon preferably is seated with ready access to master region **400**, while another surgeon or assistant may be situated near slave region **500**, which is positioned over a patient. In the embodiment of FIGS. **31A** and **31B**, master region **400** is situated laterally adjacent to slave region **500**. In addition, camera system **102** may be used with surgical robot system **100**, e.g., an endoscope that is manipulated by the assistant situated at slave region **500** may be operate and/or held in position as shown in FIG. **31B**. Camera system **102** also may include display **103** for displaying the surgical site captured by camera **102** to the surgeon in real-time. Display **103** may be mounted to master region **400**, or anywhere in proximity to master region **400** that is easily observable by the surgeon during a surgical procedure.

[0179] Referring again to FIG. **31A**, system **100** includes two hybrid telemanipulators **104** and **105**, including left hybrid telemanipulator **104** that is manipulated by the surgeon's left hand, and right hybrid telemanipulator **105** that is manipulated by the surgeon's right hand. Hybrid telemanipulators **104** and **105** may be operated simultaneously and independently from the other, e.g., by the surgeon's left and right hands. Preferably, the teleoperated remotely actuated surgical robot system **100** is optimized for use in surgical procedures.

[0180] Each hybrid telemanipulator provides input to a master-slave configuration, in which a slave unit, made of a plurality of rigid slave links and slave joints, is driven kinematically by a master unit, made of a plurality of rigid master links and master joints. For example, left hybrid

telem manipulator **104** includes master unit **401** and corresponding slave unit **501**, and right hybrid telem manipulator **105** includes master unit **402** and corresponding slave unit **502**. Master units **401** and **402** are disposed within master region **400** of system **100**, while slave units **501** and **502** are within slave region **500** of system **100**. Preferably, slave units **501** and **502** mimics the movement of the corresponding portions of master units **401** and **402**, respectively, without deviating, during operation of the device, from a remote center-of-motion, as described in further detail below.

[0181] Still referring to FIG. **31A**, teleoperated surgical instrument **106**, e.g., translational instrument interface, having end effector **107** is coupled to the distal end of slave unit **501**, and a handle is coupled to the distal end of master unit **401** such that movement applied to the handle induces a corresponding micro-movement of end-effector **107** via a processor-driven control system. For example, the control system may receive a signal indicative of movement applied at the handle by one or more sensors coupled to the handle, and perform coordinate transforms necessary to activate one or more actuators operatively coupled to end-effector **107** to replicate a corresponding movement of the end effector. Slave instrument **106** of the translational instrument interface may be removably attached to and operated by slave unit **501**, such that the translation degrees-of-freedom, e.g., left/right, upward/downward, inward/outward, are actuated by direct mechanical coupling, whereas the articulation degrees-of-freedom, e.g., pitch and yaw, the actuation degrees-of-freedom, e.g., open/close, and the rotation degree-of-freedom, e.g., pronosupination, are electromechanically replicated via sensors, actuators, and a control system as described in further detail below.

[0182] Referring now to FIGS. **32A** and **32B**, the mechanisms of exemplary remotely actuated surgical robot system **100** having hybrid telem manipulators are illustrated, in which the external covers depicted in FIG. **31** are omitted for clarity. In FIGS. **32A** and **32B**, mechanical transmission **300** is arranged to directly couple slave unit **501** with master unit **401**, such that translational macro-movement applied to the plurality of master joints of master unit **401** is replicated by corresponding respective joints of the plurality of slave joints of slave unit **501**. Likewise, mechanical transmission **300** also directly couples slave unit **502** with master unit **402**, such that translational macro-movement applied to the plurality of master joints of master unit **402** is replicated by corresponding respective joints of the plurality of slave joints of slave unit **502**. Transmission **300** illustratively includes one or more cables **301** routed via one or more pulleys from master unit **401** to slave unit **501**, and one or more cables **303** routed via one or more pulleys from master unit **402** to slave unit **502**, for controlling one of four degrees-of-freedom of slave unit **501** and **502**. Mechanical constraint **200** of master unit **401** constrains movement of master unit **401** by removing a degree-of-freedom of motion, thereby limiting movement in three translational degrees-of-freedom, e.g., left/right, upward/downward, inward/outward.

[0183] For example, one or more cables **301** may form one or more closed loops beginning at pulley **P1** coupled to master unit **401**, and extending through pulleys **P2**, **P3**, **P4**, **P5**, **P6**, tensioning system **302**, pulley **P7**, and around pulley **P8** coupled to slave unit **501**, and extending back through pulley **P7**, tensioning system **302**, pulleys **P6**, **P5**, **P4**, **P3**, **P2**, and ending at pulley **P1**. Thus, rotation of pulley **P1** clockwise or counter-clockwise causes a cable of one or more cables **301** to rotate pulley **P8**, thereby actuating a slave unit **501** in one of four degrees-of-freedoms. Mechanical constraint **200** of master unit **401**, however, constrains movement of master unit **401** by removing one degree-of-freedom of motion, thereby limiting movement of slave unit **501** to three translational degrees-of-freedom, e.g., left/right, upward/downward, inward/outward. Each of pulleys **P1**, **P2**, **P3**, **P4**, **P5**, **P6**, **P7**, and **P8** may include a number of individual pulleys corresponding to the number of degrees-of-freedom of motion actuatable of slave unit **501** by master unit **401**. Similarly, one or more cables **301** may include a number of closed cable loops corresponding to the number of degrees-of-freedom of motion actuatable of slave unit **501** by master unit **401**.

[0184] Similarly, one or more cables **303** may form one or more corresponding closed loops

beginning at pulley P9, coupled to master unit 402, and extending through tensioning system 304, pulleys P10, P11, P12, P13, P14, and around pulley P15 coupled to slave unit 502, and extending back through pulleys P14, P13, P12, P11, P10, tensioning system 304, and ending at pulley P9. In this manner, rotation of pulley P9 clockwise or counter-clockwise may cause a cable of one or more cables 303 to rotate pulley P15, thereby actuating a slave unit 502 in one of four degrees-of-freedom. Mechanical constraint 201 of master unit 402 (see FIG. 32A) likewise constrains movement of master unit 402 by removing a degree-of-freedom of motion, thereby limiting movement of slave unit 502 to three translational degrees-of-freedom, e.g., left/right, upward/downward, inward/outward. Each of pulleys P9, P10, P11, P12, P13, P14, and P15 may include a number of individual pulleys corresponding to the number of degrees-of-freedom of motion actuable of slave unit 502 by master unit 402. Similarly, one or more cables 303 may include a number of closed cable loops corresponding to the number of degrees-of-freedom of motion actuable of slave unit 502 by master unit 402.

[0185] As will be understood by a person having ordinary skill in the art, the number of pulleys P2-P7 employed to route cables 301 between pulleys P1 and P8, and the number of pulleys P10-P14 employed to route cables 303 between pulleys P9 and P15 will depend on the construction of the right and left hybrid telemanipulators, respectively.

[0186] Referring now to FIG. 33, one or more cables 301 of mechanical transmission 300 pass through tensioning system 302, and one or more cables 303 passes through tensioning system 304. Tensioning system 302 is designed to apply a predetermined tension force to cables 301, while tensioning system 304 is designed to apply a predetermined tension force to cables 303. For example, tensioning system 302 may include pulley P16 coupled to pulley P17 via tension link 305, and pulley P18 coupled to pulley P19 via tension link 306. Tension link 305 is adjustably and rotatably coupled to tension link 306 about a vertical axis running through axle 307, such that a predetermined tension force is applied to cables 301 by pulleys P16, P17, P18, and P19. In addition, tensioning system 302 may be used to calibrate mechanical transmission 300, thereby ensuring that the angles of corresponding master and slave joints are identical. Tensioning system 304 may be identical in structure to tensioning system 302.

[0187] Also in FIG. 33, pulleys P11, P12, and P13 of the mechanical transmission of the right hybrid telemanipulator are coupled to slave link 308, which is rotatably coupled to positioning system 310 via slave link 309. Positioning system 310 may be, e.g., a hydraulic device, that restricts movement of slave unit 502 with respect to slave unit 501 along a single plane. For example, the position of pulley P8 may be fixed, such that the position of P15 is moveable relative to P8 along the horizontal plane (x- and y-direction).

[0188] Referring now to FIGS. 34A and 34B, components of an exemplary master unit of system 100 is described. As master unit 401 is identical in structure to master unit 402, respectively, the description below of master unit 401 applies also to master unit 402.

[0189] Master unit 401 includes a plurality of master links, e.g., first master link 405a, second master link 405b, third master link 405c, and fourth master link, e.g., guided master link 404, interconnected by a plurality of master joints. Handle 403 is connected to a distal end of master unit 401 via guided master link 404, e.g., master rod, and includes a plurality of handle links interconnected by a plurality of handle joints for operating the hybrid telemanipulator. For example, translational macro-movement applied on handle 403 causes corresponding movement of the plurality of master joints via the plurality of master links, which is transmitted to the corresponding slave joints of slave unit 501 via mechanical transmission 300, thereby replicating the translational macro-movement at slave unit 501. Translational movement of handle 403 causes guided master link 404 to transmit motion to pulley P1 via first master link 405a, second master link 405b, and third master link 405c, thereby causing slave unit 501 to mimic the translational movement via mechanical transmission 300. First master link 405a, second master link 405b, third master link 405c, and guided master link 404 are coupled to pulley P1 via a transmission system

including, e.g., one or more toothed belts **406** routed via one or more pulleys **407**. Alternatively, the transmission system coupling pulley **P1** and the plurality of master links and joints of master unit **501** may include a system of cables and pulleys, and/or rigid transmission links.

[0190] In FIGS. **34A** and **34B**, mechanical constraint **408** on master unit **401** includes a yoke pivotally coupled to a sleeve that slides on guided master link **404** and constrains movements of the distal end of slave unit **501** in correspondence with a remote center of motion that is aligned with the incision point on a patient, e.g., the point at which a trocar passes into a patient's abdomen. For example, mechanical constraint **408** ensures that, when the hybrid telemanipulator is actuated, guided master link **404** of master unit **401** translates along longitudinal axis $\theta_{sub.1}$ so that the corresponding slave link of slave unit **501**, e.g., translational instrument interface coupled to the distal end of slave unit **501**, also translates along virtual axis $\theta_{sub.4}$ parallel to longitudinal axis $\theta_{sub.1}$ of guided master link **404** in the vicinity of the remote manipulation, as depicted in FIGS. **35A** and **35B**. In addition, mechanical constraint **408** enables guided master link **404** to rotate about second and third axes $\theta_{sub.2}$, $\theta_{sub.3}$ that are perpendicular to each other. Referring still to FIGS. **34A** and **34B**, axis $\theta_{sub.3}$ is coaxial to the axis of pulley **P1**. The plane defined by longitudinal axis $\theta_{sub.1}$ of guided master link **404** and second axis $\theta_{sub.2}$ intersects third axis $\theta_{sub.3}$ at stationary single point **409** independently of the orientation of master link **404**. This configuration allows the corresponding slave link of slave unit **501** to rotate about fifth and sixth virtual axes $\theta_{sub.5}$, $\theta_{sub.6}$ that are perpendicular to each other. Longitudinal axis $\theta_{sub.4}$ of the corresponding slave link and fifth and sixth virtual axes $\theta_{sub.5}$, $\theta_{sub.6}$ always intersect each other at virtual stationary single point **509**, e.g., the remote center-of-motion, in the vicinity of the patient incision. [0191] When surgical robot system **100** is positioned such that remote center-of-motion **509** is aligned with the patient incision, translational movement applied to handle **403** is replicated by the end-effector disposed inside the patient. Because the end-effector perfectly replicates the movement applied to handle **403**, this arrangement advantageously eliminates the fulcrum effect between the handle and end-effector, and ensures that the instrument always passes through the remote center-of-motion. Whereas in previously-known surgical robots, maintaining a fixed point of movement of the surgical instrument as it passes through the patient incision requires complex control electronics, in the system of the present invention, mechanical constraint **408** provides translational replication between master unit **401** and slave unit **501** that ensures that the instrument always passes through remote center-of-motion **509**.

[0192] Inward/outward movement of handle **403** of the embodiments of FIGS. **34A** and **34B** causes first master link **405a**, second master link **405b**, third master link **405c**, and guided master link **404** to move inward/outward along longitudinal axis $\theta_{sub.1}$ of guided master link **404**. That motion is transmitted to pulley **P1** via the plurality of master links, causing slave unit **501** to replicate the inward/outward movement about longitudinal axis $\theta_{sub.4}$ via mechanical transmission **300** and the plurality of slave links, joints, and timing belts. Similarly, movement of handle **403** upward/downward causes first master link **405a**, second master link **405b**, third master link **405c**, and guided master link **404** to rotate upward/downward about second axis $\theta_{sub.2}$. That motion is transmitted to pulley **P1** via the plurality of master links, in turn causing slave unit **501** to replicate the upward/downward movement about fifth axis $\theta_{sub.5}$ via mechanical transmission **300** and the plurality of slave links, joints, and timing belts. Finally, movement of handle **403** left/right causes first master link **405a**, second master link **405b**, third master link **405c**, and guided master link **404** to rotate left/right about third axis $\theta_{sub.3}$. That motion is transmitted to pulley **P1** via the plurality of master links, causing slave unit **501** to replicate the left/right movement about sixth axis $\theta_{sub.6}$ via mechanical transmission **300** and the plurality of slave links, joints, and timing belts.

[0193] Still referring to FIGS. **34A** and **34B**, movement applied at handle **403** of master unit **401** actuates the articulation degrees-of-freedom, e.g., pitch and yaw, the actuation degree-of-freedom, e.g., open/close, and the rotation degree-of-freedom, e.g., pronosupination, electromechanically via sensors, motors, and a control system. Master unit **401** preferably includes one or more sensors **410**

coupled to handle **403** via circuit board **411** for detecting motion of handle **403**. As will be understood, sensor **410** may be any sensor designed to detect rotational movement, such as magnetic-based rotational sensors that includes a magnet on one side and a sensor on another side to measure rotation by measuring angle and position. Circuit board **411** is coupled to a control system for generating signals indicative of the rotation measured by sensor **410** and transmitting the signals to one or more motors coupled to slave unit **501**, which may reproduce movements applied on handle **403** to the end effector. For example, electrical cables may extend from handle **403** to the control system, e.g., a unit containing control electronics, and additional electrical cables may extend from the control system to the one or more motors coupled to slave unit **501**.

[0194] Actuation of trigger **412** of handle **403** generates a signal that is transmitted via the control system to the motors coupled to slave unit **501**, thereby causing actuation of a translation transmission system of the translational instrument interface coupled to slave unit **501**, in turn causing actuation of the end-effector of the translational instrument interface to open/close.

[0195] Handle **403** also may include ball **413** designed to be easily gripped by the surgeon and which aligns the surgeon's wrist with master unit **401**. Ball **413** may be rotatable about handle axis $\theta_{\text{sub.7}}$, such that the rotation of ball **413** is detected by a sensor that generates and transmits a signal via the control system to a motor coupled to slave unit **501**. The signal received from the control system at the slave unit causes rotation of the translational instrument interface coupled to slave unit **501**, thus rotating the end-effector of the translational instrument interface in the pronosupination degree-of-freedom.

[0196] Handle **403** also is rotatable about handle axis $\theta_{\text{sub.8}}$, such that the rotation about handle axis $\theta_{\text{sub.8}}$ is detected by a sensor, which generates and transmits a signal via the control system to the motors of slave unit **501**. That signal causes actuation of the translation transmission system of the translational instrument interface coupled to slave unit **501**, which in turn causes movement of the end-effector of the translational instrument interface in the yaw degree-of-freedom. In addition, handle **403** may be rotatable about handle axis $\theta_{\text{sub.9}}$, such that the rotation of handle **403** about handle axis $\theta_{\text{sub.9}}$ is detected by a sensor, which generates and transmits a signal via the control system to the motors of slave unit **501**. That signal causes actuation of the translation transmission system of the translational instrument interface coupled to slave unit **501**, which causes movement of the end-effector of the translational instrument interface in the pitch degree-of-freedom.

[0197] Referring now to FIGS. **34C** and **34D**, alternative embodiments of the handle of master unit **401** are described. In FIG. **34C**, handle **403'** is rotatable about handle axis $\theta_{\text{sub.7}}$, handle axis $\theta_{\text{sub.8}}$, and handle axis $\theta_{\text{sub.9}}$, such that rotation of handle **403'** about the handle axes is detected by one or more sensors **410**, which generate and transmit a signal via the control system to motors of slave unit **501**. That signal actuates the translation transmission system of the translational instrument interface coupled to slave unit **501**, causing movement of the end-effector of the translational instrument interface in the pronosupination, yaw, and pitch degrees-of-freedom, respectively.

[0198] Similarly, handle **403''** of FIG. **34D** is rotatable about handle axis $\theta_{\text{sub.7}}$, handle axis $\theta_{\text{sub.8}}$, and handle axis $\theta_{\text{sub.9}}$, such that rotation of handle **403''** about the handle axes is detected by one or more sensors **410**, which generates and transmits a signal via the control system to the one or more motors coupled to slave unit **501**. That signal actuates the translation transmission system of the translational instrument interface coupled to slave unit **501**, causing movement of the end-effector of the translational instrument interface in the pronosupination, yaw, and pitch degrees-of-freedom, respectively.

[0199] Referring now to FIGS. **35A** and **35B**, an exemplary slave unit of system **100** is described. As slave unit **501** is identical in structure to slave unit **502**, respectively, the description below of slave unit **501** applies also to slave unit **502**.

[0200] As described above, master unit **401** includes a plurality of master links interconnected by a plurality of master joints. Slave unit **501** includes a corresponding plurality of slave links, e.g., first

slave link **505a**, second slave link **505b**, third slave link **505c**, and fourth slave link, e.g., translational instrument interface **503**, interconnected by a plurality of slave joints, such that a direct mechanical coupling is formed by the plurality of slave links and corresponding plurality of slave joints of slave unit **501**, which is identical to the kinematic model formed by the corresponding plurality of master links and corresponding plurality of master joints of master unit **401**. For example, first slave link **505a** always remains parallel to first master link **405a**, second slave link **505b** always remains parallel to second master link **405b**, third slave link **505c** always remains parallel to third master link **405c**, and translational instrument interface **503** always remains parallel to guided master link **404** during operation of the hybrid telemanipulator. Thus, each translational macro-movement applied to the plurality of master joints of master unit **401** is replicated by a corresponding respective joint of the plurality of slave joints of slave unit **501** via mechanical transmission **300** and the plurality of slave links.

[0201] In FIGS. **35A** and **35B**, translational instrument interface **503** is coupled to distal end **504** of slave unit **501**. Translational movement of handle **403** is transmitted to pulley **P9** via mechanical transmission **300**. More specifically, translational actuation of handle **403** causes pulley **P9** to transmit motion to end-effector **512** via first slave link **505a**, second slave link **505b**, third slave link **505c**, and translational instrument interface **503**, thereby causing slave unit **501** to replicate the translational movement. First slave link **505a**, second slave link **505b**, third slave link **505c**, and translational instrument interface **503** are coupled to pulley **P9** via a transmission system including, e.g., one or more timing belts **506** routed via one or more pulleys **507**. Thus, each of the four pulleys of **P9** is operatively coupled to and controls movement of first slave link **505a**, second slave link **505b**, third slave link **505c**, and translational instrument interface **503**. Alternatively, the transmission system coupling pulley **P9** and the plurality of slave links and joints of slave unit **501** may include a system of cables and pulleys, and/or rigid transmission links.

[0202] Mechanical constraint **408** of master unit **401** ensures that, when the hybrid telemanipulator is in operation, first slave link **505a**, second slave link **505b**, third slave link **505c**, and translational instrument interface **503** always rotate about virtual stationary point **509**. For example, end-effector **512** of translational instrument interface **503** coupled to slave unit **501** always translates along longitudinal axis $\theta_{\text{sub.4}}$ corresponding to the longitudinal axis $\theta_{\text{sub.1}}$ of master link **404** in the vicinity of the remote manipulation. In addition, mechanical constraint **408** allows end-effector **512** to rotate about fifth and a sixth virtual axis $\theta_{\text{sub.5}}$, $\theta_{\text{sub.6}}$ that are perpendicular to each other. Longitudinal axis $\theta_{\text{sub.4}}$ of translational instrument interface **503** coupled to slave unit **501**, and fifth and sixth virtual axes $\theta_{\text{sub.5}}$, $\theta_{\text{sub.6}}$ always intersect each other at virtual stationary single point **509** in the vicinity of the remote manipulation. During a minimally invasive surgical procedure, virtual stationary point **509** is aligned with the surgical incision point, reducing trauma to the patient and improving cosmetic outcomes of the surgery.

[0203] Movement of handle **403** in the inward/outward directions causes end effector **512** coupled to slave unit **501** to replicate the inward/outward movement about longitudinal axis $\theta_{\text{sub.4}}$ via mechanical transmission **300** and the transmission system coupling pulley **P9** and the plurality of slave links and joints of slave unit **501**. Movement of handle **403** upward/downward causes end effector **512** coupled to slave unit **501** to replicate the upward/downward movement about longitudinal axis $\theta_{\text{sub.5}}$ via mechanical transmission **300** and the transmission system coupling pulley **P9** and the plurality of slave links and joints of slave unit **501**. Movement of handle **403** left/right causes end effector **512** coupled to slave unit **501** to replicate the left/right movement about longitudinal axis $\theta_{\text{sub.6}}$ via mechanical transmission **300** and the transmission system coupling pulley **P9** and the plurality of slave links and joints of slave unit **501**.

[0204] In addition, movement applied at handle **403** of master unit **401** actuates the articulation degrees-of-freedom, e.g., pitch and yaw, the actuation degree-of-freedom, e.g., open/close, and the rotation degree-of-freedom, e.g., pronosupination of the end-effector of translational instrument interface **503**, electromechanically via sensors, motors, and a control system. Translational

instrument interface **503** may be constructed as described in U.S. Patent Publication No.

2018/0353252 to Chassot, assigned to the assignee of the instant application, the entire contents of which are incorporated by reference herein. For example, translational instrument interface **503** includes slave hub **510** and surgical instrument **511**. Slave hub **510** may be affixed to distal end **504** of slave unit **501**. Surgical instrument **511** includes end-effector **512** disposed at the distal end of the shaft of surgical instrument **511**, and may be removably coupled to slave hub **510**. A sterile interface may be positioned between slave hub **510** and surgical instrument **511**. In addition, translational instrument interface **503** includes a translation transmission system that extends from one or more motors positioned within slave hub **510** to the components of end-effector **512**. For example, end-effector **512** includes a plurality of end-effector links interconnected by a plurality of end-effector joints coupled to the translation transmission system of translational instrument interface **503**, such that actuation of the translation transmission system by the one or more motors causes movement of end-effector **512** via the plurality of end-effector links and joints.

[0205] Further details regarding the components and operation of slave hub **510** are described with respect to FIGS. **36A** and **36B**. Slave hub **510** of translational instrument interface **503** affixed to slave unit **501** includes one or more motors, e.g., first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d**, operatively coupled, e.g., via electrical wiring, with the control system via circuit board **602**. Motors **601a-601d** receive signals indicative of measured movements and trigger actuation of handle **403**, as measured by one or more sensors **410** coupled to handle **403**. Those signals are processed by the control system, which in turn provides signals to the motors that actuate translational instrument interface **503** to thereby replicate the micro-movements corresponding to those input at handle **403**. First motor **601a**, second motor **601b**, and third motor **601c** are coupled directly to translation transmission system **603** of translational instrument interface **503** for actuating end-effector **512** in the open/close, pitch, and yaw degrees-of-freedom. Translation transmission system **603** includes a plurality of transmission elements, e.g., cables and/or lead screws, such that each of the plurality of transmission elements are coupled to first motor **601a**, second motor **601b**, and third motor **601c** at one end, and to the first, second, and third end-effector links at the opposite end, to move the end-effector in the open/close, pitch, and yaw degrees-of-freedom. Translation transmission system **603** may include a plurality of lead screws and/or closed cable loops. Fourth motor **601d** actuates rotation of slave instrument **503** via pronosupination timing belt **513**. As will be understood by a person having ordinary skill in the art, slave hub **510** may include any combination of motors **601a-601d**, e.g., only the one or more motors for actuating end-effector **512** in the open/close degree-of-freedom and the motor for rotating end-effector **512** in the pronosupination degree-of-freedom when a non-articulated instrument is used.

[0206] Circuit board **602** also may include one or more sensors designed to detect undesired movement of translational instrument interface **503**, and electrically communicate with first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d** to resist such undesired movement.

[0207] In accordance with one aspect of the present invention, the control system may identify the kinematics of end-effector **512** of translational instrument interface **503** by reading out identifier element **516**, e.g., RFID token integrated with the instrument, as shown in FIG. **36C**, wherein the RFID token contains information on the kinematic configuration of the instrument. In particular, the control system, based on the information read from identifier element **516**, may configure operation of the one or more motors that interface with translational instrument interface **503** to operate differently (e.g. to turn simultaneously clockwise or one clockwise and the other counterclockwise) to cause actuations of the end-effector elements. For example, in FIG. **36D**, a forceps-type end-effector having parallel-serial instrument kinematics is described. For this configuration, first motor **601a** may be operatively coupled to a first link of end-effector **512'**, e.g., a first blade, via transmission element **514a** of the translation transmission system, such that first

motor **601a** causes the first link of end-effector **512'** to move outward/inward. Second motor **601b** may be operatively coupled to a second link of end-effector **512'**, e.g., second blade, via transmission element **514b** of the translation transmission system, such that second motor **601b** causes the second link of end-effector **512'** to move outward/inward. Thus, the control system may instruct first motor **601a** to move the first link of end-effector **512'** outward via transmission element **514a**, while simultaneously instructing second motor **601b** to move the second link of end-effector **512'** outward via transmission element **514b**, thereby causing end-effector **512'** to open based on actuation of trigger **412** of handle **403**. Conversely, the control system may instruct first motor **601a** to move the first link of end-effector **512'** inward via transmission element **514a**, while simultaneously instructing second motor **601b** to move the second link of end-effector **512'** inward via transmission element **514b**, thereby causing end-effector **512'** to close based on actuation of trigger **412** of handle **403**. Therefore, first motor **601a** and second motor **601b** may cause end-effector **512'** to move in the open/close degree-of-freedom.

[0208] The control system may instruct first motor **601a** to move the first link of end-effector **512'** outward via transmission element **514a**, while simultaneously instructing second motor **601b** to move the second link of end-effector **512'** inward via transmission element **514b**, thereby causing end-effector **512'** to pitch upward based on rotation of handle **403** about handle axis $\theta_{\text{sub.9}}$.

Conversely, the control system may instruct first motor **601a** to move the first link of end-effector **512'** inward via transmission element **514a**, while simultaneously instructing second motor **601b** to move the second link of end-effector **512'** outward via transmission element **514b**, thereby causing end-effector **512'** to pitch downward based on rotation of handle **403** about handle axis $\theta_{\text{sub.9}}$. Therefore, first motor **601a** and second motor **601b** may cause end-effector **512'** to move in the pitch degree-of-freedom.

[0209] Third motor **601c** may be operatively coupled to a third link of end-effector **512'** via transmission element **514c** of the translation transmission system such that third motor **601c** causes end-effector **512'** to move in the yaw degree-of-freedom based on rotation of handle **403** about handle axis $\theta_{\text{sub.8}}$. Fourth motor **601d** may be operatively coupled to first motor **601a**, second motor **601b**, third motor **601c**, and surgical instrument **511** via a rotatable pronosupination timing belt **513** such that fourth motor **601d** causes first motor **601a**, second motor **601b**, third motor **601c**, and surgical instrument **511**, and thereby end-effector **512'**, to rotate in the pronosupination degree-of-freedom based on rotation of ball **413** of handle **403**.

[0210] Referring now to FIG. 36E, an end-effector having serial-serial instrument kinematics is described. For example, first motor **601a** may be operatively coupled to a first link of end-effector **512''** via transmission element **515a** of the translation transmission system such that first motor **601a** causes end-effector **512''** to move in the open/close degree-of-freedom based on actuation of trigger **412** of handle **403**. Second motor **601b** may be operatively coupled to a second link of end-effector **512''** via transmission element **515b** of the translation transmission system such that second motor **601b** causes end-effector **512''** to move in the pitch degree-of-freedom based on rotation of handle **403** about handle axis $\theta_{\text{sub.9}}$. Third motor **601c** may be operatively coupled to a third link of end-effector **512''** via transmission element **515c** of the translation transmission system such that third motor **601c** causes end-effector **512''** to move in the yaw degree-of-freedom based on rotation of handle **403** about handle axis θ_s . Fourth motor **601d** may be operatively coupled to first motor **601a**, second motor **601b**, third motor **601c**, and surgical instrument **511** via a rotatable pronosupination timing belt **513** such that fourth motor **601d** causes first motor **601a**, second motor **601b**, third motor **601c**, and surgical instrument **511**, and thereby end-effector **512''**, to rotate in the pronosupination degree-of-freedom based on rotation of ball **413** of handle **403**.

[0211] In accordance with one aspect of the invention, the control system may read information stored on identifier element **516**, e.g., an RFID token, that is integrated with the instrument to identify the kinematics of end-effector **512** of translational instrument interface **503**, as outlined in method steps **700** enumerated in FIG. 37. At step **701**, the user selects a surgical instrument having

an end-effector to be used with the hybrid telemanipulator. For example, the surgical instrument may have an end-effector having parallel-serial instrument kinematics as shown in FIG. 36D or serial-serial instrument kinematics as shown in FIG. 36E. The surgical instrument then may be coupled to the slave unit of the hybrid telemanipulator. At step 702, the control system detects information for the kinematic configuration of the selected end-effector. For example, the control system may read out an RFID token integrated with surgical instrument 511, which contains information on the kinematic configuration of the selected end-effector, e.g., whether the selected end-effector has parallel-serial instrument kinematics or serial-serial instrument kinematics. The RFID token may be, for example, an inductively-read microchip that contains identification information that may be scanned by a reader disposed on the slave hub and operatively coupled to the control system. Alternatively, the function of identifier element 516 may be provided by, e.g., an optical tag such as a bar code, QR code, Datamatrix, Aztec code, or Semacode, disposed on the surgical instrument 511 that is read by the slave hub. If the surgical instrument has not yet been coupled to the slave unit of the hybrid telemanipulator, surgical instrument may be coupled to the slave unit of the hybrid telemanipulator after step 702.

[0212] At step 703, the control system identifies the kinematics of the selected end-effector based on the information detected at step 702 to determine which type of end-effector is coupled to the slave unit of the hybrid telemanipulator. At step 704, the control system adjusts its parameters based on the identity of the selected end-effector so that the hybrid telemanipulator may be properly actuated. For example, if the end-effector has parallel-serial instrument kinematics, the control system will include parameters that instruct first motor 601a and second motor 601b to simultaneously actuate the first and second end-effector links to move the end-effector in the open/close and pitch degrees-of-freedom as described above. If the end-effector has serial-serial instrument kinematics, the control system will include parameters that instruct first motor 601a to actuate the end-effector in the open/close degree-of-freedom, and second motor 601b to actuate the end-effector in the pitch degree-of-freedom as described above.

[0213] Referring to FIG. 38, an alternative exemplary embodiment of a remotely actuated surgical robot system wherein all degrees-of-freedom are controlled electromechanically is described. Although all seven of the degrees-of-freedom, e.g., inward/outward, upward/downward, left/right, yaw, pitch, open/close, and pronosupination, are controlled electromechanically via a system of sensors, motors, and a control system, system 800 retains a mechanical constraint element as described above at the master unit, thereby creating a single virtual stationary point, e.g., remote center-of-motion, at the slave unit. Therefore, system 800 does not require coordinate transform and complex control system to align slave unit 1001 with an incision. The mechanical constraint and the corresponding remote-center-of-motion ensure that this design, is much simpler and safer than when using generic robotic arms.

[0214] Referring now to FIG. 39, master unit 901 is constructed similarly to master unit 401 of FIGS. 34A and 34B, except that instead of a plurality of cables and pulleys of the mechanical transmission coupled to pulley P1, master unit 901 includes one or more sensors, e.g., sensor 902a, sensor 902b, sensor 902c, and sensor 902d, operatively coupled to each of the four pulleys of pulley P1. Sensors 902a-902d measure rotational movement by measuring angle and position of pulley P1 in response to movement applied to handle 903 of master unit 901 via a plurality of master links, joints, and cables. Each of the four sensors measures movement of a joint of master unit 901 via each of the four pulleys of pulley P1, thereby measuring movement of master unit 901 in four degrees-of-motion. However, the mechanical constraint constrains movement of master unit 901 by removing one degree-of-freedom of motion, thereby resulting in movement of slave unit 1001 in three degrees-of-freedom of motion, e.g., inward/outward, upward/downward, and left/right.

[0215] Handle 903 is constructed similarly to handle 403 of FIGS. 34A and 34B. For example, handle 903 includes one or more sensors 410 and circuit board 411, such that micro movements

applied at handle **903** may be transmitted to the end-effector of slave unit **1001** via one or more sensors **410** and the one or more motors coupled to the end-effector of slave unit **1001** to move the end-effector in the open/close, pitch, yaw, and pronosupination degrees-of-freedom.

[0216] Regarding transmission of macro-movements, sensor **902a**, sensor **902b**, sensor **902c**, and sensor **902d** generate signals indicative of the measured rotation of pulley **P1** by the respective sensors, and transmit the signals to one or more motors coupled to slave unit **1001** via a control system, to thereby replicate the translational macro-movements applied at handle **903** coupled to master unit **901**. For example, electrical cables may extend from master unit **901** to the control system, e.g., unit containing control electronics, and additional electrical cables may extend from the control system to the one or more motors coupled to slave unit **1001**.

[0217] With respect to FIGS. **40A** and **40B**, slave unit **1001** is constructed similarly to slave unit **501** of FIGS. **35A** and **35B**. For example, slave unit **1001** includes first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d** operatively coupled to the end-effector of slave unit **1001**, such that micro-movements applied at handle **903** may be transmitted to the end-effector of slave unit **1001** via one or more sensors **410**, and first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d** to move the end-effector in the open/close, pitch, yaw, and pronosupination degrees-of-freedom. Slave unit **1001** differs from slave unit **501** in that instead of a plurality of cables and pulleys of the mechanical transmission coupled to pulley **P8**, slave unit **1001** includes one or more motors, e.g., first motor **1002a**, second motor **1002b**, third motor **1002c**, and fourth motor **1002d**, operatively coupled to each of the four pulleys of pulley **P8**. The one or more motors are coupled to a circuit board for receiving signals indicative of the measured rotation of pulley **P1** by sensor **902a**, sensor **902b**, sensor **902c**, and sensor **902d** in response to movement applied to handle **903** of master unit **901**, to thereby actuate pulley **P8** to replicate the translational macro-movements applied on handle **903** coupled to master unit **901** at slave unit **1001** via a plurality of slave links, joints, timing belts, and/or a system of cables and pulleys. For example, first motor **1002a** is operatively coupled to and controls movement of first slave link **505a**, second motor **1002b** is operatively coupled and controls movement of second slave link **505b**, third motor **1002c** is operatively coupled to and controls movement of third slave link **505c**, and fourth motor **1002d** is operatively coupled to and controls movement of translational instrument interface **503** via pulley **P8** and the plurality of slave joints, timing belts, and/or system of cables and pulleys.

[0218] As the mechanical constraint of master unit **901** constrains movement of master unit **901** to movement in three degrees-of-freedom, e.g., inward/outward, upward/downward, and left/right, movement of first slave link **505a**, second slave link **505b**, third slave link **505c**, and translational instrument interface **503** of slave unit **1001** by first motor **1002a**, second motor **1002b**, third motor **1002c**, and fourth motor **1002d**, respectively, is constrained to movement in three degrees-of-freedom, e.g., inward/outward, upward/downward, and left/right, about virtual stationary point **1005**, e.g., remote center-of-motion.

[0219] Slave unit **1001** may include temporary incision pointer **1004** which points to virtual stationary point **1005**, e.g., remote center-of-motion, created by the mechanical restraint at master unit **1001**, such that virtual stationary point **1005** may be brought in coincidence with the surgical incision point, reducing trauma to the patient and improving cosmetic outcomes of the surgery. Temporary incision pointer **1004** is removably coupled to a joint of slave unit **1001** such that temporary incision pointer **1004** points to virtual stationary point **1005**, and may be removed prior to operation of surgical robot system **800**.

[0220] Referring now to FIGS. **40C** and **40D**, an alternative exemplary embodiment of the incision pointer is provided that is utilized with the systems shown in and described in FIGS. **1-30** herein. Like incision pointer **1004**, incision pointer **1004'** may be removably coupled to a joint of slave unit **1001** (e.g., the distal end of link **63** described above) such that temporary incision pointer **1004** points to a virtual center-of-motion, e.g., virtual stationary point **1005**, thereby identifying the

remote center-of-motion of the surgical instrument. For example, incision point **1004'** may be removable coupled to the slave console using structures including, but not limited to, magnets, friction forces, Velcro surfaces, matching geometries, hooks, etc. Incision pointer **1004'** may be made of ferritic stainless steel, and illustratively include magnetic head **1006** at a proximal end of the incision pointer for magnetically coupling with a corresponding surface at the slave console (e.g., the receptacle at the joint of slave unit **1001**).

[0221] As shown in FIG. **40C**, magnetic head **1006** may have a convex spherical surface with its center aligned with the distal tip of incision pointer **1004'**. Accordingly, the receptacle at the joint of slave unit **1001** may have a corresponding concave spherical surface for engaging with the convex spherical surface of incision pointer **1004'**. As will be understood by a person having ordinary skill in the art, magnetic head **1006** of incision pointer **1004'** may have a concave spherical surface, and the receptacle at the joint of slave unit **1001** may have a convex spherical surface. Incision point **1004'** may function with or without a sterile drape installed over slave unit **1001**. FIG. **40E** shows incision pointer **1004'** inserted within trocar **1007** such that the distal tip of incision pointer **1004'** is aligned with the body wall of the patient, e.g., virtual stationary point **1005**, about which the instrument will rotate. Incision pointer **1004'** may be removed from the receptacle at the joint of slave unit **1001** after the surgical instrument is inserted, prior to operation of the surgical robot system.

[0222] In accordance with one aspect of the present invention, incision pointer **1004'** may be removably coupled to the joint of slave unit **1001**, e.g., at link **63**, and a clinician may move the joint of slave unit **1001**, and accordingly, the neighboring plurality of slave links and slave joints, until the distal tip of incision pointer **1004'** is aligned with and points to a desired location on the patient's body, e.g., the incision site on the patient's body. Once the joint of slave unit **1001** and incision pointer **1004'** are pointing at the desired location, the clinician may set the desired location as virtual stationary point **1005** via the control system based on the alignment of the joint of slave unit **1001**. Accordingly, all movement of the surgical instrument by the slave console during operation of the surgical robot system will always rotate about virtual stationary point **1005**, even without a mechanical constraint at the master console as described in further detail below with reference to FIG. **43**. In this manner, the controller of the system ensures that the surgical instrument does not cause translational movements away from the virtual stationary point **1005** for safe surgery through a trocar. The aligned link (e.g., link **63**) preferably always points at the surgical site (e.g., opening through the trocar) during the surgery such that movement of the surgical instrument is restricted about the virtual stationary point.

[0223] Accordingly, the controller of the system may execute instructions to set virtual stationary point **1005** based on the alignment of link **63** and the desired location on the patient's body at the surgical site, such that movement of the surgical instrument is restricted about virtual stationary point **1005** to maintain alignment of link **63** (and its longitudinal axis $\theta_{sub.5}$) with the incision site during the surgery.

[0224] Referring now to FIGS. **41A** and **41B**, alternative embodiments of the control system of the surgical robot system are described. Control system **1100** of FIG. **41A**, which may be integrated with system **100**, includes non-transitory computer readable media, e.g., memory **1101**, having instructions stored thereon that, when executed by processor **1102** of control system **1100** allow operation of the hybrid telemanipulators. In addition, control system **1100** may communicate with identifier element reader **517** of slave unit **501**, either wirelessly or using an electric cable, so that memory **1101** may store the identity of the kinematic configuration of the end-effector read from identifier element **516**, such that the instructions, when executed by processor **1102**, cause the motors for controlling the end-effector in the open/close and pitch degrees-of-freedom to behave in accordance to the type of end-effector selected. Control system **1100** is electrically coupled, either wirelessly or using an electric cable, to the circuit boards of master unit **401** and, thereby, to one or more sensors **410** for receiving signals indicative of micro movements applied at handle **403**. In

addition, control system **1100** is electrically coupled, either wirelessly or using an electric cable, to the circuit boards of slave unit **501** and, thereby, to first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d** for actuating the micro movements of the end-effector, e.g., in the open/close, pitch, yaw, and pronosupination degrees-of-freedom.

[0225] Control system **1110** of FIG. **41B**, which may be integrated with system **800**, includes non-transitory computer readable media, e.g., memory **1111**, having instructions stored thereon that, when executed by processor **1112** of control system **1110** allow operation of the hybrid telemanipulators. In addition, control system **1110** may communicate with identifier element reader **517** of slave unit **1001**, either wirelessly or using an electric cable, and memory **1111** may store the identity of the kinematic configuration of the end-effector read from identifier element **516**, such that the instructions, when executed by processor **1112**, cause the motors for controlling the end-effector in the open/close and pitch degrees-of-freedom to behave in accordance to the type of end-effector selected. Control system **1110** is electrically coupled, either wirelessly or using an electric cable, to the circuit boards of master unit **901** and, thereby, to one or more sensors **410** for receiving signals indicative of micro movements applied at handle **903**, and to sensor **902a**, sensor **902b**, sensor **902c**, and sensor **902d** for receiving signals indicative of macro movements applied at handle **903**. In addition, control system **1110** is electrically coupled, either wirelessly or using an electric cable, to the circuit boards of slave unit **1001** and, thereby, to first motor **601a**, second motor **601b**, third motor **601c**, and fourth motor **601d** for actuating the micro movements of the end-effector, e.g., in the open/close, pitch, yaw, and pronosupination degrees-of-freedom, and to first motor **1002a**, second motor **1002b**, third motor **1002c**, and fourth motor **1002d** for actuating the macro-movements of the end-effector, e.g., in the inward/outward, upward/downward, and left/right degrees-of-freedom.

[0226] Referring now to FIGS. **42A** and **42B**, alternative applications of the principles of the present invention may be applied to alternative telemanipulator designs. For example, a telemanipulator constructed as described in U.S. Pat. No. 9,696,700 to Beira, depicted in FIG. **42A**, may be modified to include handles and translational instrument interfaces for electromechanically controlling the micro movements of the end-effector, e.g., open/close, pitch, yaw, and pronosupination degrees-of-freedom, while the translational macro movements of the end effector, e.g., upward/downward, inward/outward, and left/right degrees-of-freedom, are controlled mechanically by a mechanical transmission system. Remotely actuated surgical robot system **1200** includes master unit **1201** directly mechanically coupled to slave unit **1202**, handle **1203** coupled to master unit **1201**, translational instrument interface **1204** coupled to slave unit **1202**, and mechanical constraint **1205**. Handle **1203** may be constructed similarly to handle **403** of FIGS. **34A** and **34B**, and translational instrument interface **1204** likewise may be constructed similarly to translational instrument interface **503** of FIGS. **35A** and **35B**. For example, handle **1203** includes one or more sensors, such that micro-movements applied at handle **1203** may be transmitted to the end-effector of translational instrument interface **1204** via the one or more sensors and one or more motors coupled to the end-effector of slave unit **1202** to move the end-effector in the open/close, pitch, yaw, and pronosupination degrees-of-freedom. Accordingly, the translational macro-movements applied at handle **1201** will be replicated by translational instrument interface **1204** in three degrees-of-freedom, e.g., inward/outward, upward/downward, and left/right, due to mechanical constraint **1203**. Alternatively, remotely actuated surgical robot system **1200** may have seven degrees-of-freedom actuated electromechanically.

[0227] With respect to FIG. **42B**, an alternative telemanipulator is described. For example, a telemanipulator constructed as described in U.S. Patent Pub. No. 2017/0245954 to Beira, may be modified to include handles and translational instrument interfaces for electromechanically controlling the micro-movements of the end-effector, e.g., open/close, pitch, yaw, and pronosupination degrees-of-freedom, while the translational macro-movements of the end effector, e.g., upward/downward, inward/outward, and left/right degrees-of-freedom, are controlled

mechanically by a mechanical transmission system. Remotely actuated surgical robot system **1210** includes master unit **1211** mechanically coupled to slave unit **1212**, handle **1213** coupled to master unit **1211**, translational instrument interface **1214** coupled to slave unit **1212**, and mechanical constraint **1215**. Handle **1213** is constructed similarly to handle **403** of FIGS. **34A** and **34B**, and translational instrument interface **1214** is constructed similarly to translational instrument interface **503** of FIGS. **35A** and **35B**. For example, handle **1213** includes one or more sensors, such that micro-movements applied at handle **1213** may be transmitted to the end-effector of translational instrument interface **1214** via the one or more sensors and one or more motors coupled to the end-effector of slave unit **1212** to move the end-effector in the open/close, pitch, yaw, and pronosupination degrees-of-freedom. Accordingly, the translational macro-movements applied at handle **1213** will be replicated by the end-effector of translational instrument interface **1214** in three degrees-of-freedom, e.g., inward/outward, upward/downward, and left/right, due to mechanical constraint **1213**. Alternatively, remotely actuated surgical robot system **1210** may have seven degrees-of-freedom actuated electromechanically.

[0228] Referring now to FIG. **43**, another exemplary master console constructed in accordance with the principles of the present invention is provided. Master console **20'** is constructed similar to master console **20** of FIG. **2A**, except that the master telemanipulator of master console **20'** does not include a mechanical constraint designed to constrain movement of at least one master link of the plurality of master links as described above. For example, a master telemanipulator of master console **20'** includes a base portion having telescoping bases **1008** and **1009** for adjusting the vertical height of the master telemanipulator, and base cap **1010** fixed atop telescoping bases **1008** and **1009** and rotatably coupled to link **26** via joint **25**.

[0229] Unlike master console **20** of FIG. **2A**, the master telemanipulator of master console **20'** includes link **1014** coupled to link **1012** via joint **1013**, and link **1016** coupled to link **1014** via link **1015** and further coupled to handle portion **1018** via joint **1017**. As shown in FIG. **43**, none of links **1014** or **1016** passes through link **1012**. Instead, once the virtual stationary point, e.g., remote center-of-motion, is set by the control system as described above with reference to FIG. **40C-40E**, movements applied to the master telemanipulator by the surgeon will be made by slave links and slave joints of the slave console in a corresponding manner about the virtual stationary point.

[0230] Advantageously, master console **20** permits the surgeon to approach the handle grip of handle portion **1018** from top-down (instead of bottom-up or horizontally as with other surgical robots). In addition, master console **20** shares the same orientation as the surgeon's arms such that the base of the master arms is positioned on the side of the surgeon's body, and not in front in the area of the central pillar of master console **20**. As the sterile master arms of master console **20** are positioned further from the ground in this configuration, sterility of master console **20** is better maintained during use of the surgical system. Moreover, this configuration reduces the depth of the surgeon console, thereby saving valuable operating room floor space.

[0231] Further, as shown in FIG. **43**, master console **20'** may include master arm break release button **1019** and height adjustment button **1020**. For example, actuation of master arm break release button **1019** will permit the user to readjust the plurality of master links and master joints until the master telemanipulator is in a desired configured for use by the surgeon, and actuation of height adjustment button **1020** will permit the user to adjust the vertical height, e.g., up or down, of the master telemanipulator via telescoping bases **1008** and **1009**.

[0232] While various illustrative embodiments of the invention are described above, it will be apparent to one skilled in the art that various changes and modifications may be made therein without departing from the invention. The appended claims are intended to cover all such changes and modifications that fall within the true scope of the invention.

Claims

- 1.** A system for remote manipulation to perform surgery, the system comprising: a patient console comprising a plurality of patient links coupled to a base; a surgical instrument coupled to the patient console, a distal region of the surgical instrument configured to be inserted into a patient at a surgical site for performing robotic surgery; and a controller configured to execute instructions to: cause, in a surgical mode, at least one of the plurality of patient links to move responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery; and cause the patient console to transition from the surgical mode to a laparoscopic mode where the plurality of patient links are retracted away from the patient while the base of the patient console remains stationary to expose the surgical site to permit a surgeon to perform non-robotic surgery at the surgical site without interference from the plurality of patient links.
- 2.** The system of claim 1, wherein the controller is further configured to execute instructions to: determine that the surgical instrument has been removed from the patient at the surgical site; and only cause the patient console to transition from the surgical mode to the laparoscopic mode if the surgical instrument has been removed.
- 3.** The system of claim 2, wherein the controller determines that the surgical instrument has been removed from the patient by determining that the surgical instrument has been decoupled from the patient console.
- 4.** The system of claim 1, wherein the controller causes the patient console to transition from the surgical mode to the laparoscopic mode responsive to user input received at the patient console.
- 5.** The system of claim 1, wherein the handle is removably coupled to the surgeon console such that the handle is sterile during the surgery and sterilizable while removed for additional surgeries.
- 6.** The system of claim 5, further comprising a sterile drape interface having a ring defining an opening, wherein the handle, when coupled to the surgeon console, holds the ring in place so that the sterile drape of the sterile drape interface covers portions of the surgeon console.
- 7.** The system of claim 1, wherein the controller is configured to execute instructions to cause, in the surgical mode, the at least one of the plurality of patient links to move responsive to movement applied at the handle of the surgeon console at a scaled degree.
- 8.** The system of claim 7, wherein the controller is configured to execute instructions to cause, in the surgical mode, scaled micro movements at the surgical instrument in a micro degree-of-freedom responsive to corresponding movement applied at the handle of the surgeon console.
- 9.** The system of claim 8, wherein micro movements applied at the surgical instrument are independently scalable for each of the micro degrees-of-freedom such that the scaled micro movements in the micro degree-of-freedom are at a different scale than second scaled micro movement at the surgical instrument in a second micro degree-of-freedom.
- 10.** The system of claim 1, wherein the surgeon console comprises a clutch configured to prevent micro-movements at the surgical instrument responsive to micro-movements applied at the handle of the surgeon console when the clutch is actuated.
- 11.** A method for remotely performing a surgery, the method comprising: coupling a surgical instrument to a patient console comprising a plurality of patient links coupled to a base; inserting a distal region of the surgical instrument into a patient at a surgical site for performing robotic surgery; moving, in a surgical mode, at least one of the plurality of patient links responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery; and transitioning the patient console from the surgical mode to a laparoscopic mode where the plurality of patient links are retracted away from the patient while the base of the patient console remains stationary to expose the surgical site to permit a surgeon to perform non-robotic surgery at the surgical site without interference from the plurality of patient links.
- 12.** The method of claim 11, further comprising: determining that the surgical instrument has been

removed from the patient at the surgical site, wherein transitioning the patient console from the surgical mode to the laparoscopic mode only occurs if the surgical instrument has been removed.

13. The method of claim 12, wherein determining that the surgical instrument has been removed from the patient at the surgical site comprises determining that the surgical instrument has been decoupled from the patient console.

14. The method of claim 11, further comprising receiving user input at the patient console, wherein transitioning the patient console from the surgical mode to the laparoscopic mode is responsive to the user input received at the patient console.

15. The method of claim 11, wherein the handle is removably coupled to the surgeon console, the method further comprising removing the handle for sterilization in-between surgeries.

16. A system for remote manipulation to perform surgery, the system comprising: a patient console comprising an alignment joint and a plurality of patient links coupled to a base; a surgical instrument coupled to the patient console, a distal region of the surgical instrument configured to be inserted into a patient at a surgical site for performing robotic surgery; and a controller configured to execute instructions to: set a virtual center-of-motion based on an alignment of the alignment joint and the surgical site; and cause at least one of the plurality of patient links to move responsive to movement applied at a handle of a surgeon console operatively coupled to the patient console to thereby move the surgical instrument to perform the robotic surgery, wherein movement of the surgical instrument is restricted about the virtual remote center-of-motion to maintain alignment of the patient joint with the surgical site during the surgery.

17. The system of claim 16, further comprising an incision pointer configured to be removably coupled to the alignment joint to permit alignment of the alignment joint and the surgical site.

18. The system of claim 17, wherein the incision pointer is configured to be removably coupled to the alignment joint via a magnetic attachment.

19. The system of claim 16, wherein the virtual center-of-motion is set based on the alignment of the alignment joint and a trocar positioned within the patient at the surgical site.

20. The system of claim 16, wherein the handle is removably coupled to the surgeon console such that the handle is sterile during the surgery and sterilizable while removed for additional surgeries.
